

MTH302 – Business Mathematics & Statistics

Important MCQ's and SEQ's from Handouts – Midterm

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Chap 2

→ Basic arithmetic operations

+

-

x

÷

exponents

$$(4)^2 = 16$$

$$(4)^{-1/2} = \frac{1}{(4)^{1/2}} = \frac{1}{2} = 0.5$$

→ Excel 2002 XP is latest version of this software

→ columns →
rows ↓

→ Intersection of row & column is cell.

→ All cells starting from leftmost corner

→ Range starts to rightmost corner in last row

→ Range specified by starting cell, a colon & ending cell

- Excel has 4 types of operations
 - Arithmetic operators $+$ $-$ $\times$ $/$ $\%$
 - Comparison $=$
 - Text concatenation $=$ combine text
 - Reference operator has $:$ $:$ $:$

- excel formula for $\times$ is similar to $+$

Chap 3

- Three types of employees
 - Regular monthly salary
 - part time hourly based
 - payment on per piece basis.

- Gross earning / salary.

Basic

Allowances

- Gross salary may include

Basic

House rent

Conveyance allowance

Utilities allowance

- Structure of allowances
 - House rent: 45%.
 - Conveyance = 2.5%.
 - Utilities allowance = 2.5%.

→ Example

If basic salary is 10,000 Rs.

$$\text{House rent} = 0.45 \times 10,000 = 4500 \text{ Rs.}$$

$$\text{Conveyance allowance} = 0.025 \times 10,000 = 250$$

$$\text{Utilities allowance} = 0.025 \times 10,000 = 250$$

$$\begin{aligned} \text{total allowances} &= 4500 + 250 + 250 \\ &= 5000 \text{ Rs.} \end{aligned}$$

→ Provident fund

By law $1/11^{\text{th}}$ of Basic salary per month is deducted by company from gross earnings of employee

An equal amount $1/11^{\text{th}}$ of basic salary per month

There is investment of $2/11^{\text{th}}$ of basic salary on behalf of employee in provident fund

Company can invest savings
in govt approved securities
such as defense saving certificates

Interest earned on investments
in provident fund is credited
to the account of employees
in proportion to their
share in provident fund

→ Example

Basic salary	10k
Allowances	5k

Employee contribution to provident *

$$\text{fund} = \frac{1}{11} \times 10k = 909.1 \text{ Rs}$$

Company contribution to provident

$$\text{fund} = \frac{1}{11} \times 10k = 909.1 \text{ Rs}$$

Total savings of employee in
Provident fund = $909.1 + 909.1 =$
 1818.2 Rs

→ Gratuity fund
for benefit of employees, at time of retirement.
By law $1/11\text{th}$ of basic salary per month is contributed by company to Gratuity fund to account of employee

There is saving of $1/11\text{th}$ of basic salary on behalf of employee in gratuity fund in govt approved securities i.e. defense saving certificates.

Interest earned on investment is credited to employee account

→ Example

Basic salary 10k
allowance 5k.

Total savings of employee in gratuity fund $= 1/11\text{th} \times 10\text{k} = 909.1$

→ Leaves

Casual 18 days per year
Earned " " "
Sick 12 " "

→ Example

Basic 10k
Allowance 5k

→ Gross salary = Basic + allowance

→ Cost of casual leaves per year
 $= (18 / 22 \times 12) \times 15k \times 12$
 $= 12,272.7 \text{ Rs.}$

→ Social Charges

comprise leaves group insurance
& medical that is 5% of
gross salary. employees

→ children's education
club membership leave fare
assistance such benefits
may be 5-8%.

→ Leaves are 13.2% of gross
salary

* Total social charges =
 leaves as from gross salary + medical /
 group insurance + children
 education club membership
 leave fare assistance

* Total social charges
 $= 18.2 + 5 + 5.8 = 29\%$

29% social charges are quite common.

→ Example

Basic salary 10k

Allowance 5k

Group insurance/medical =
 $0.05 \times 15k = 750$

Other social benefits

$0.058 \times 15k = 870$

Leave cost = $0.182 \times 15k$
 $= 2730$

Total social charges

$= 2730 + 750 + 870$

4350 Rs

→ Gross remuneration / gross wage
pay or salary
includes

Basic salary 6k

House rent 45%

conveyance 2.5%

utilities 2.5%

Provident 1/11

Gratuity 1/11

Leaves

group medical insurance

other social charges

All this ↑ economic security
of employees.

→ Summary

Basic salary 6k

Allowances 50% of basic salary

Gratuity 9.09% of basic salary

Provident fund 9.09% " "

Social charges 29%

→ Converting fraction to %.

$$\text{Percent} = \text{fraction} \times 100$$

→ convert 0.1 to %.

Ans 10%.

→ common fraction's example is

$$20\% = \frac{20}{100} = 0.2$$

→ Decimal fraction's example is

$$20\% = \frac{20}{100} = 0.2$$

→ % = Base $\times$ Rate

→ What % is 20% of 120

$$= \frac{20}{100} \times 120$$

$$= 24$$

$$\rightarrow 6\% \text{ of } 40 = \frac{6}{100} \times 40 = 2.4$$

→ Base = Percentage / Rate

→ example

$$\text{Base} = \frac{96}{24/100} = 400 \quad *$$

Lesson 4

→ Excel formula is with in bracket

→ (Excel formula = \$B\$93 * 0.45)

→ Average (Arithmetic mean)
= Sum/N

→ total salary = SUM(B93:B96)

→ Gratuity = ROUND((1/11) * \$B\$93, 0)

→ Auto sum symbol is Σ

SUMIF function to create total
value for range

→ DSUM

Add num in column of list or database that match conditions

→ Syntax

DSUM (database, field, criteria)

→ Database is range of cells that make up list or database

→ Field indicates the column is used in function

→ weighted average

Some elements of set carry more importance than others.

Lecture 5

Change = final - initial value

% change = $\left(\frac{\text{Change}}{\text{initial value}} \right) \times 100\%$

→ Example

initial value = 1000

final = 2500

change = 1500

$$\% \text{ change} = \frac{1500}{1000} \times 100 = 150\%$$

→

Monday's sale = 1000

Next day sale = 2500

Next day sale as % =

$$= \frac{2500}{1000} \times 100 = 250\%$$

2 and a half times

$$\Rightarrow = D_{13} / D_{12} \times 100$$

→ Original fruit 15 kg

final fruit = 3 kg

Change = -12

→ $\% \text{ Change} = \text{Final} - \text{original}$

$$\% \text{ change} = \frac{-12}{15} \times 100 = -80\%$$

$$\Rightarrow \% \text{ change} = \frac{\text{change}}{\text{original}} \times 100$$

→ Example
 Currently earning 5000
 ↑ 3% 2% 1% in successive
 years.

$$= 5000 (1 + 3\%) (1 + 2\%) (1 + 1\%)$$

$$= 5000 \times 1.03 \times 1.02 \times 1.01$$

$$= 5306$$

→ Example

Invested 1 lac

Rate of return for each year

4% , 8% , -10% , 9%

How much at end of 4th year?

① 4% = 1 lac $\times (1 + \frac{4}{100})$ = Ans.
 = 104000

② 8% = 104000 $\times (1 + \frac{8}{100})$ = 112320

③ -10% = 112320 $\times (1 - \frac{10}{100})$ = 101088

④ 9% = 101088 $\times (1 + \frac{9}{100})$
 = 110186

Call 11 = Power (0.1 + (57/100))^{0.1}
Call 11 = Power (1.57 (1 + (57/100))^{0.1})

Chap 6

Example

→ interest rate 9% to 7%

% ↓ = 2

↑ interest rate on loan 7% to 9%

% ↑ = 2

↓ in interest rate = 7 - 9 = -2%

% ↓ = $\frac{\text{final}}{\text{initial}} \times 100$

$$= -2/9 \times 100 = -22.2\%$$

↑ in interest rate = 9 - 7 = 2%

% increase = $2/9 \times 100 = 22.2\%$

→ Stock represents company assets

& earnings

Share equity or stock

all means same thing

→ stock yield
represented in %age form.

$$= \frac{\text{Annual dividend payment}}{\text{Stock current share price}}$$

→ Earning per share EPS.

$$\frac{\text{Total profit}}{\text{No of shares}}$$

Example 18 earning \$ 200 million shares = \$5^{per share}

→ Price earning ratio PER

$$= \frac{\text{Market value per share}}{\text{Earning per share}}$$

Example company trading at \$43 a share
earning \$1.95 per share.

→ Outstanding shares
stock currently held by
investors, restricted shares
owned by officers & insiders
& those held by public.

→ Net current asset value per share
$$NCAVPS = \frac{\text{Current assets} - \text{total liabilities}}{\text{\# of shares outstanding}}$$

→ Current assets
expected to be converted into
cash within one year

Cash accounts receivable inventory
marketable securities, prepaid
expenses & other liquid assets
that can be readily converted to
cash

→ Liabilities
company's legal debts or
obligations that arise during
course of business operations

→ Market value
Value at which investors
buy or sell share of stock
at a given time

→ Face value / Par value
original cost of a share
of stock which is shown
on certificate

It bears no relationship to its
market price.

→ Dividend
company distributes a part
of profit it earns as dividend
dividend is %age of share's
face value

example

if face value is 10
50% dividend will mean a
dividend of Rs 5 per share

Buying shares

Example If you buy 100 shares at
62.50 per share with 2%
commission

Calculate total cost

$$62.50 \times 100$$

$$= 6250$$

$$2\% \text{ commission } 6250 \times \frac{2}{100} = 125$$

$$\text{Total cost} = 6250 + 125$$

$$= 6375$$

$$\rightarrow \text{Return on investment} = \frac{\text{total gain}}{\text{total cost}}$$

Example

100 shares at 52.25

Sold 1 year later at 68

1% commission rate of buying & selling

10% dividend per share
face value 10 Rs.

Return on investment = ?

$$\text{Bought} = 100 \times 52.25 = 5225$$

$$\text{Total cost} = 5225 \times \frac{101}{100} = 5277.25$$

$$\text{Sold} = 68 \times 100 = 6800$$

$$6800 \times \frac{99}{100} = 6732$$

Net gain = minus above both ans give 1454.75

Total gain = Net gain + dividend that is 10% $\left(\frac{100 \times 10}{100} \right)$

→ Example

List price = 2200

Discount rate = 15%

Discount

$$2200 \times 0.15 = 330$$



net cost price = List price - discount

$$= 2200 - 330 = \text{Ans.}$$

* Simple interest = Price $\times$ rate of interest $\times$ time
in yr

→

Compound amount S

P price

r rate

n number of period

*

$$S = P \left(1 + \frac{r}{100} \right)^n$$

*

Compound interest = S - P

Lecture 7

Chap 4

Annuity
series of ^{fixed} payments over
a period of time required
from you or paid to you at
a specified frequency

use to build retirement
income

to -1 to save for child's
edu, trust fund

most common payment frequencies are
yearly

semi-annually

quarterly

monthly

* future value ordinary annuity

$$FV = C \times \left[\frac{(1 + i)^n - 1}{i} \right]$$

example

$$\$1000 \times \left[\frac{(1 + 0.05)^5 - 1}{0.05} \right]$$

* PV Present value ordinary annuity

$$PV = \frac{C \left[1 - (1+i)^{-n} \right]}{i}$$

* $1 - (1+i)^{-n} / i$ is discount factor

* FV ordinary annuity

$$FV = C \left[\frac{(1+i)^n - 1}{i} \right]$$

* Accumulate value

$$S = \frac{C \left[(1+i)^n - 1 \right]}{i}$$

* Accumulated factor $(1+i)^n / i$

Accumulated value

payment per period $\times$

accumulation factor for n periods

$\rightarrow$ 10% of 110 is

$$110 \times \frac{10}{100} = 11$$

$\rightarrow$ Present value = 121 at end of year 2 is 100

→ When future value is known and present worth rate of interest calculation are made is called Discount factor rate.

→ Present worth of future payments is called discounted value

* Formula of discount factor

$$= (1 - 1/(1+i)^n) / i$$

* Algebraic operation are numbers & variables

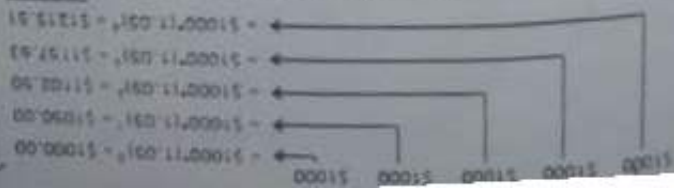
Components are separated by

algebraic operations separated by

algebraic factors substituted

into numerical & literal coefficients

Future Value of an Ordinary Annuity = \$5525.63



Since we have to add the future value of each payment, you may have noticed that if you have an annuity with many cash flows, it would take a long time to calculate all the future values and then add them together. Fortunately, mathematics provides a formula that serves as a short-cut for finding the accumulated value of all cash flows received from an annuity.

$$FV_{\text{Ordinary Annuity}} = C \cdot \left[\frac{(1+i)^n - 1}{i} \right]$$

C = Payment per period or amount of annuity
i = interest rate
n = number of payments

$\frac{(1+i)^n - 1}{i}$ is called accumulation factor for n periods

Accumulated value of n period = payment per period $\times$ accumulation factor for n periods

if we were to use the above formula for Example 1 above, this is the result:

$$FV_{\text{Ordinary Annuity}} = \$1000 \cdot \left[\frac{(1+0.05)^5 - 1}{0.05} \right] = \$5525.63$$

Note that the \$0.01 difference between \$5,525.64 and \$5,525.63 is due to a rounding error in the first calculation. Each of the values of the first calculation must be rounded to the nearest penny - the more you have to round numbers in a calculation the more likely rounding errors will occur. So, the above formula not only provides a short-cut to finding FV of an ordinary annuity but also gives a more accurate result.

Calculating the Present Value or Discounted Value of an Annuity

If you would like to determine today's value of a series of future payments, you need to use formula that calculates the present value of an ordinary annuity.

For Example 2, we'll use the same annuity cash flow schedule as we did in Example 1. To find the total discounted value, we need to take the present value of each future payment and add it to the present value of the first payment. For Example 1, we did in Example 1, add the cash flows together.

$\frac{\$1000}{(1.05)^1}$	= \$952.38
$\frac{\$1000}{(1.05)^2}$	= 907.03
$\frac{\$1000}{(1.05)^3}$	= 863.84
$\frac{\$1000}{(1.05)^4}$	= 822.70
$\frac{\$1000}{(1.05)^5}$	= 783.53

Present Value of an Ordinary Annuity = \$4329.48

Again, calculating and adding all these values will take a considerable amount of time, especially if we expect many future payments. As such, there is a mathematical shortcut we can use for ordinary annuity.

$$PV_{\text{Ordinary Annuity}} = C \cdot \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

C = Cash flow per period

i = interest rate

n = number of payments

$(1 - (1+i)^{-n}) / i$ is called discount factor for n periods.

Thus Discounted value of n period = payment per period × discount factor for n period

The formula provides us with the PV in a few easy steps. Here is the calculation of the annuity represented in the diagram for Example 2.

$$\begin{aligned}
 PV_{\text{Ordinary Annuity}} &= \$1000 \cdot \left[\frac{1 - (1+0.05)^{-5}}{0.05} \right] \\
 &= \$1000 \cdot [4.33] \\
 &= \$4329.48
 \end{aligned}$$

NOTATIONS

The following notations are used in calculations of Annuity:

R = Amount of annuity

N = Number of payments

i = Interest rate per conversion period

S = Accumulated value

A = Discounted or present worth of an annuity

ACCUMULATED VALUE

The accumulated value S of an annuity is the total payments made including the interest. The formula for Accumulated Value S is as follows:

$$S = r \cdot ((1+i)^n - 1) / i$$

★ Accumulation factor for n payments = $((1+i)^n - 1) / i$

It may be seen that:

★ Accumulated value = Payment per period × Accumulation factor for n payments

LECTURE 8

Compound Interest

Calculate returns from investments

Annuities

Excel Functions

OBJECTIVES

The objectives of the lecture are to learn about:

- Review of lecture 7
- Compound Interest
- Calculate returns from investments
- Annuities
- Excel Functions

CUMIPMT

Returns the cumulative interest paid on a loan between start_period and end_period if the function is not available, and returns the #NAME? error. Install and load the Analysis ToolPak add-in.

The syntax is as follows:

CUMIPMT(rate,nper,pv,start_period,end_period,type)

Rate: interest rate
 Nper: total number of payment periods
 Pv: present value
 Start_period: first period in the calculation
 End_period: last period in the calculation
 Type: timing of the payment

Type	Timing
0 (zero)	Payment at the end of the period
1	Payment at the beginning of the period

CUMIPMT-EXAMPLE

Following is an example of CUMIPMT function. In this example, in the first case the objective is to find total interest paid in the second year of payments for periods 13 to 24. Please note there are 12 periods per year. The second case is for the first payment period.

In the first formula, the Annual interest rate 9% is cell A2 (not shown here). The Years of the loan are given in cell A3. The Present value is in cell A4. For the Start period the value 13 was entered. For the End period, the value 24 has been specified. The value of Type is 0, which means that the payment will be at the end of the period. Please note that the annual interest is first divided by 12 to arrive at monthly interest. Then the Years of the loan are multiplied by 12 to get total number of months in the Term of the loan. The answer is (-11135.23).

In the second formula, which gives Interest paid in a single payment in the first month 1 was specified as the Start period. For the End period also the value 1 was entered. This is because only 1 period is under study. All other inputs were the same. The answer is (-937.50).

Date	Description
9%	Annual interest rate
30	Years of the loan
125,000	Present value
=CUMPRINC(A2/12,A3*12,A4,13,24,0) The total principal paid in the second year of payments, periods 13 through 24 (-934,107.1)	
=CUMPRINC(A2/12,A3*12,A4,1,1) The principal paid in a single payment in the first month (-68,278.27)	

CUMPRINC
The CUMPRINC function returns the cumulative principal paid on a loan between two periods.
The syntax is as under:

CUMPRINC(rate,nper,pv,start_period,end_period,type)

Rate: interest rate.
Nper: total number of payment periods.
Pv: present value.
Start_period: period in the calculation. Payment.
End_period: last period in the calculation.
Type: timing of the payment (0 or 1 as above).

CUMPRINC EXAMPLE

Following is an example of CUMPRINC function. In this example, in the first case the objective is to find the total principal paid in the second year of payments, periods 13 through 24. Please note there are 12 periods per year. The second case is for the principal paid in a single payment in the first month.
In the first formula, the interest rate per annum 9% is in cell A2 (not shown here). The term in years (30) is given in cell A3. The Present value is in cell A4. For the start period the value 13 was entered. For the End period, the value 24 has been specified. The value of Type is 0, which means that the payment will be at the end of the period. Please note that the interest is first divided by 12 to arrive at monthly interest. Then the years of loan are multiplied by 12 to get total number of months in the term of the loan. The answer is (-934,107.1).

In the second formula, which gives the principal paid in a single payment in the first month 1 was specified as the start period. For the end period also the value 1 was entered. This is because only 1 period is under study. All other inputs were the same. The answer is (-68,278.27).

EXAMPLE

Date	Description
9.00%	Interest rate per annum
30	Term in years
125,000	Present value
=CUMPRINC(A2/12,A3*12,A4,13,24,0) The total principal paid in the second year of payments, periods 13 through 24 (-934,107.1)	
=CUMPRINC(A2/12,A3*12,A4,1,1,0) The principal paid in a single payment in the first month (-68,278.27)	

EFFECT

Returns the effective annual interest rate. As you see there are only two inputs, namely, the nominal interest rate and the number of compounding periods per year.
EFFECT(nominal_rate,npery)

Nominal_rate: nominal interest rate
Npery: number of compounding periods per year

EFFECT-EXAMPLE

Rate: Nominal rate = 5.25% in cell A2. Nper: 4 in cell A3. The answer is 0.053543 or 5.3543%. You should round off the value to 2 decimals: 5.35%.

5.25% Nominal interest rate
4 Number of compounding periods per year
=EFFECT(A2,A3)
(Effective interest rate with the terms above (0.053543 or 5.3543 percent))

FV
Returns the future value of an investment. There are 5 inputs, namely: Rate the interest rate, Nper number of periods, Pmt payment per period, Pv present value and Type.

=FV(rate,nper,pmt,pv,type)
Rate: interest rate per period
Nper: total number of payment periods
Pmt: payment made each period
Pv: present value, or the lump-sum amount
Type: number 0 or 1 due

FV-EXAMPLE 1

In the formula, there are 5 inputs, namely: Rate 6% in cell A2 as the interest rate, 10 as Nper number of periods in cell A3, -200 (notice the minus sign) as Pmt payment per period in cell A4, -500 (notice the minus sign) as Pv present value in cell A4 and 1 as Type in cell A6. The answer is (2581.40).

Cell	Description	EXAMPLE 1
B2	Annual interest rate	
B3	Number of payments	
B4	Amount of the payment	
B5	Present value	
B6	Payment is due at the beginning of the period	

=FV(B2:B6, A3, A4, A5, A6)
Future value of an investment with the above terms (2581.40)
=FV(rate,nper,pmt,pv,type)

FV-EXAMPLE 2

In the formula, there are 4 inputs, namely, Rate 11% in cell A2 as the interest rate, 35 as Nper number of periods in cell A3, -2000 (notice the minus sign) as Pmt payment per period in cell A4, 1 as Type in cell A5. The value of Pv was omitted by entering a blank for the value (note the double commas",,"). The answer is (82846.25).

EXAMPLE 2

12% Annual interest rate
 12 Number of payments
 -1000 Amount of the payment
 =FV(A2/12, A3, A4)
 Future value of an investment
 with the above terms
 (12,682.50)
 FV(rate,nper,pmt,pv,type)

FV-EXAMPLE 3

In the formula, there are 4 inputs, namely, Rate 11% in cell A2 as the interest rate, 35 as Nper number of periods in cell A3, -2000 (notice the minus sign) as Pmt payment per period in cell A4, 1 as Type in cell A5. The value of Pv was omitted by entering a blank for the value (note the double commas",,"). The answer is (82846.25).

EXAMPLE 3

11% Annual interest rate
 35 Number of payments
 -2000 Amount of the payment
 1 Payment is due at the beginning
 of the period
 =FV(A2/12, A3, A4, A5)
 Future value of an investment with the
 above terms (82,846.25)
 FV(rate,nper,pmt,pv,type)

Returns the future value of an initial principal after applying a series of compound interest rates.
FVSCHEDULE(principal, schedule)
 principal: present value
 schedule: an array of interest rates to apply

FVSCHEDULE-EXAMPLE

In this example, the Principal is 1. The compound rates (0.09, 0.11, 0.1) are given within curly brackets. The answer is (1.33089).
FVSCHEDULE(principal, schedule)
 =**FVSCHEDULE**(1, {0.09, 0.11, 0.1})
 Future value of 1 with compound interest rates of 0.09, 0.11, 0.1 (1.33089)

IPMT

Returns the interest payment for an investment for a given period.
IPMT(rate, per, nper, pv, fv, type)
 Rate: interest rate per period
 Per: period to find the interest
 Nper: total number of payment periods
 Pv: present value, or the lump-sum amount
 Fv: future value, or a cash balance
 Type: number 0 or 1

ISPMT

Calculates the interest paid during a specific period of an investment.
ISPMT(rate, per, nper, pv)
 Rate: interest rate
 Per: period
 Nper: total number of payment periods
 Pv: present value. For a loan, pv is the loan amount

NOMINAL

Returns the annual nominal interest rate.
NOMINAL(effect_rate, npery)
 Effect_rate: effective interest rate
 Npery: number of compounding periods per year

NPER

Returns the number of periods for an investment.
NPER(rate, pmt, pv, fv, type)
 Rate: the interest rate per period
 Pmt: payment made each period
 Pv: present value, or the lump-sum amount
 Fv: future value, or a cash balance
 Type: number 0 or 1 (due)

NPV

Returns the net present value of an investment based on a series of periodic cash flows and a discount rate. Its syntax is
NPV(rate, value1, value2, ...)
 Rate: is the rate of discount over the length of one period.
 Value1, value2, ... are 1 to 29 arguments representing the payments and income.

PMT

Returns the periodic payment for an annuity.
PMT(rate, nper, pv, fv, type)
 Rate: interest rate
 Nper: total number of payments
 Pv: present value
 Fv: future value

Type: number 0 (zero) or 1

PPMT

Returns the payment on the principal for an investment for a given period.

PPMT(rate,nper,pv,fv,type)

Rate: interest rate per period

Nper: total number of payment periods

Pv: the present value

Fv: future value (0)

Type: the number 0 or 1 (due)

PV

Returns the present value of an investment.

PV(rate,nper,pmt,fv,type)

Rate: interest rate per period

Nper: total number of payment periods in an annuity

Pmt: payment made each period and cannot change over the life of the annuity

Fv: future value, or a cash balance

Type: number 0 or 1 and indicates when payments are due

RATE

Returns the interest rate per period of an annuity.

RATE(nper,pmt,pv,fv,type,guess)

Nper: total number of payment periods

Pmt: payment made each period

Pv: present value

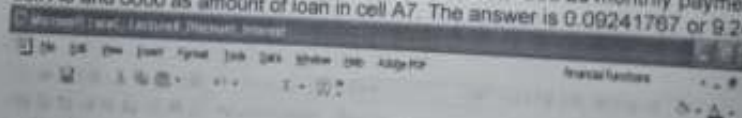
Fv: future value, or a cash balance (0)

Type: number 0 or 1 (due)

Guess: (10%)

RATE-EXAMPLE

Three inputs are specified, 4 as years of loan in cell A5, -200 as monthly payment in cell A6 and 8000 as amount of loan in cell A7. The answer is 0.09241767 or 9.24%.



RATE		RATE(nper,pmt,pv,fv,type,guess)	
Data	Description		
4	Years of the loan		
-200	Monthly payment		
8000	Amount of the loan		
=RATE(A5*12;A6;A7)			
0.092		Annual rate of the loan	(1%)
		(0.09241767 or 9.24%)	

QUESTIONS

What questions would you be asked about this material? There are many important questions:
When do we use Matrices?
Typical applications?
What is a Matrix?
What are Matrix operations?
What are Matrix Functions?
There are many applications of matrices in business and industry especially where large amounts of data are processed daily.

TYPICAL APPLICATIONS

Practical questions in modern business and economic management can be answered with the help of matrix representation in

- Economics
- Network Analysis
- Transport Networks
- Communication
- Linear Programming
- Analysis of GDP
- Computer graphics

WHAT IS A MATRIX?

A matrix is a rectangular array of numbers. The plural of matrix is matrices. Matrices are usually represented with capital letters such as A, B, C . For example

$$A = \begin{pmatrix} 1 & 9 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 47 & 52 & 70 & 56 \\ 52 & 33 & 28 & 46 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 4 \\ 2 & 3 \\ 3 & 5 \end{pmatrix}$$

The numbers in a matrix are often arranged in a meaningful way. For example, the order for school clothing in September is illustrated in the table, as well as in the corresponding matrix.

		Size			
	Youth	S	M	L	XL
Sweat Pants	5	10	34	40	12
Sweat Shirts	18	25	28	21	7
Shirts	19	10	45	38	16
T-shirts	27	7	10	24	10

The data in the above table can be entered in the shape of a matrix as follows:

$$\begin{bmatrix} 0 & 10 & 34 & 40 & 12 \\ 18 & 25 & 29 & 21 & 7 \\ 19 & 13 & 48 & 35 & 9 \\ 27 & 7 & 10 & 24 & 14 \end{bmatrix}$$

DIMENSION

Dimension or Order of a Matrix = Number of Rows x Number of Columns

Example

Matrix T has dimensions of 2×3 or the order of matrix T is 2×3 . 'x' is just the notation, it does not mean to multiply both of them.

$$T = \begin{bmatrix} 8 & 2 & -1 \\ -4 & 0 & 7 \end{bmatrix}$$

$\begin{matrix} \nearrow & \uparrow & \nwarrow \\ \text{col 1} & \text{col 2} & \text{col 3} \end{matrix}$

→ row

ROW, COLUMN AND SQUARE MATRIX

Suppose $n = 1, 2, 3, 4, \dots$

A matrix with dimensions $1 \times n$ is referred to as a row matrix.

For example, matrix A is a 1×4 row matrix.

A matrix with dimensions $n \times 1$ is referred to as a column matrix.

For example, matrix B in the middle is a 2×1 column matrix.

A matrix with dimensions $n \times n$ is referred to as a square matrix.

For example, matrix C is a 3×3 square matrix.

$$A = [12 \quad 17 \quad 10 \quad 9] \quad B = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 & 7 \\ 6 & 8 & -1 \\ -5 & 0 & 4 \end{bmatrix}$$

IDENTITY MATRIX

An identity matrix is a square matrix with 1's on the main diagonal from the upper left to the lower right and 0's off the main diagonal. An identity matrix is denoted as I .

Some examples of identity matrices are shown below. The subscript indicates the

size of the identity matrix. For example, I_n represents an identity matrix with dimensions $n \times n$.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

MULTIPLICATIVE IDENTITY

With real numbers, the number 1 is referred to as a multiplicative identity because it has the unique property that the product of a real number and 1 is that real number. In other words, 1 is called a multiplicative identity because for any real number n , $1 \cdot n = n$ and $n \cdot 1 = n$. With matrices, the identity matrix shares the same unique property as the number 1. In other words, a 2×2 identity matrix is a multiplicative inverse because for any 2×2 matrix A, $I_2 \cdot A = A$ and $A \cdot I_2 = A$.

Example

6. Matrix and its Inverse: Types of matrix

Given the 2×2 matrix, $A = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$

$$I_2 \cdot A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$$

$$A \cdot I_2 = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$$

Then

$$r_1c_1 = 1(2) + 0(-3) = 2$$

$$r_2c_1 = 0(2) + 1(-3) = -3$$

$$r_1c_2 = 1(-1) + 0(4) = -1$$

$$r_2c_2 = 0(-1) + 1(4) = 4$$

$$r_1c_3 = 1(-1) + 0(4) = -1$$

$$r_2c_3 = 0(-1) + 1(4) = 4$$

$$r_1c_4 = 1(0) + 0(1) = 0$$

$$r_2c_4 = 0(0) + 1(1) = 1$$

$$r_3c_1 = 0(2) + 1(-3) = -3$$

$$r_3c_2 = 0(-1) + 1(4) = 4$$

where 'r' is for row and 'c' is for column.

LECTURE 10 MATRICES

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 9
- Matrices

EXAMPLE 1

An athletic clothing company manufactures T-shirts and sweat shirts in four different sizes, small, medium, large, and x-large. The company supplies two major universities, the U of R and the U of S. The tables below show September's clothing order for each university.

University of S's September Clothing Order

	S	M	L	XL
T-shirts	100	300	500	300
sweat shirts	150	400	450	250

University of R's September Clothing Order

	S	M	L	XL
T-shirts	60	250	400	250
sweat shirts	100	200	350	200

Matrix Representation

The above information can be given by two matrices S and R as shown below.

$$S = \begin{bmatrix} 100 & 300 & 500 & 300 \\ 150 & 400 & 450 & 250 \end{bmatrix}$$

$$R = \begin{bmatrix} 60 & 250 & 400 & 250 \\ 100 & 200 & 350 & 200 \end{bmatrix}$$

MATRIX OPERATIONS

The matrix operations can be summarized as under:

- Organize and interpret data using matrices
- Use matrices in business applications
- Add and subtract two matrices
- Multiply a matrix by a scalar
- Multiply two matrices
- Interpret the meaning of the elements within a product matrix

PRODUCTION

The clothing company production in preparation for the universities' September orders is shown by the table and corresponding matrix P below.

	S	M	L	XL
T-shirts	300	700	900	500
sweat shirts	300	700	900	500

$$P = \begin{bmatrix} 300 & 700 & 900 & 500 \\ 300 & 700 & 900 & 500 \end{bmatrix}$$

ADDITION AND SUBTRACTION OF MATRICES

The sum or difference of two matrices is calculated by adding or subtracting the corresponding elements of the matrices. To add or subtract matrices, they must have the same dimensions.

PRODUCTION REQUIREMENT

Since the U of S ordered 100 small T-shirts and the U of R ordered 60, then altogether 160 small T-shirts are required to supply both universities. Thus, to calculate the total number of T-shirts and sweat shirts required to supply both universities, add the corresponding elements of the two order matrices as shown below.

$$\begin{bmatrix} 100 & 300 & 500 & 300 \\ 150 & 400 & 450 & 250 \end{bmatrix} + \begin{bmatrix} 60 & 250 & 400 & 250 \\ 100 & 200 & 350 & 220 \end{bmatrix}$$

$$= \begin{bmatrix} 160 & 550 & 900 & 550 \\ 250 & 600 & 800 & 470 \end{bmatrix}$$

OVERPRODUCTION

Since the company produced 300 small T-shirts and the received orders for only 160 small T-shirts, then the company produced 140 small T-shirts too many. Thus, to determine the company's over-production, subtract the corresponding elements of the total order matrix from the production matrix as shown below.

$$\begin{bmatrix} 300 & 700 & 900 & 500 \\ 300 & 700 & 900 & 500 \end{bmatrix} - \begin{bmatrix} 160 & 550 & 900 & 550 \\ 250 & 600 & 800 & 470 \end{bmatrix} = \begin{bmatrix} 140 & 150 & 0 & -50 \\ 50 & 100 & 100 & 30 \end{bmatrix}$$

MULTIPLY A MATRIX BY A SCALAR

Given a matrix A and a number c , the scalar multiplication cA is computed by multiplying the scalar c by every element of A . For example:

$$2 \begin{bmatrix} 1 & 8 & -3 \\ 4 & -2 & 5 \end{bmatrix} = \begin{bmatrix} 2 \times 1 & 2 \times 8 & 2 \times -3 \\ 2 \times 4 & 2 \times -2 & 2 \times 5 \end{bmatrix} = \begin{bmatrix} 2 & 16 & -6 \\ 8 & -4 & 10 \end{bmatrix}$$

MULTIPLICATION OF MATRICES

To understand the reasoning behind the definition of matrix multiplication, let us consider the following example. Competing Companies, A and B, sell juice in 591 mL, 1 L, and 2 L plastic bottles at prices of Rs. 1.60, Rs. 2.30 and Rs. 3.10, respectively. The table below summarizes the sales for the two companies during the month of July.

	991ml	1L	2L
Company A	20,000	5,500	10,600
Company B	18,250	7,000	11,000

What is total revenue of Company A?

What is total revenue of Company B?

Matrices may be used to illustrate the above information.

As shown at the right, the sales can be written as a column matrix, P , and the a 2×3 matrix, S , the selling prices can be written as a column matrix, R , the total revenue for each company can be expressed as a column matrix, R .

$$S = \begin{bmatrix} 20,000 & 5,500 & 10,600 \\ 18,250 & 7,000 & 11,000 \end{bmatrix} \quad P = \begin{bmatrix} 1.60 \\ 2.30 \\ 3.10 \end{bmatrix} \quad R = \begin{bmatrix} 77,510 \\ 79,400 \end{bmatrix}$$

Since revenue is calculated by multiplying the number of sales by the selling price, the total revenue for each company is found by taking the product of the sales matrix and the price matrix.

$$\begin{bmatrix} 20,000 & 5,500 & 10,600 \\ 18,250 & 7,000 & 11,000 \end{bmatrix} \times \begin{bmatrix} 1.60 \\ 2.30 \\ 3.10 \end{bmatrix} = \begin{bmatrix} 77,510 \\ 79,400 \end{bmatrix}$$

Consider how the first row of matrix S and the single column P lead to the first entry of R .

$$\begin{aligned} & \left(\begin{bmatrix} 20,000 & 5,500 & 10,600 \\ 18,250 & 7,000 & 11,000 \end{bmatrix} \times \begin{bmatrix} 1.60 \\ 2.30 \\ 3.10 \end{bmatrix} \right) \\ & \Rightarrow 20,000(1.60) + 5,500(2.30) + 10,600(3.10) \Rightarrow \begin{bmatrix} 77,510 \\ 79,400 \end{bmatrix} \\ & \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ & \text{Product of} \quad \text{Product of} \quad \text{Product of} \\ & \text{First Entries} \quad \text{Second Entries} \quad \text{Third Entries} \end{aligned}$$

With the above in mind, we define the product of a row and a column to be the number obtained by multiplying corresponding entries (first by first, second by second, and so on) and adding the results.

MULTIPLICATION RULES

If matrix A is a $m \times n$ matrix and matrix B is a $n \times p$ matrix, then the product AB is the $m \times p$ matrix whose entry in the i -th row and the j -th column is the product of the i -th row of matrix A and the j -th row of matrix B .

The product of a row and a column is the number obtained by multiplying corresponding elements (first by first, second by second, and so on).

To multiply matrices, the number of columns of A must equal the number of rows of B.

For example

Given the matrices below, decide if the indicated product exists. And, if the product exists, determine the dimensions of the product matrix.

$$A = \begin{bmatrix} -5 & 0 & 4 \\ 12 & -3 & 10 \\ 7 & 15 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 \\ 9 & -1 \\ -5 & 7 \end{bmatrix}$$

MULTIPLICATION CHECKS

The table below gives a summary whether it is possible to multiply two matrices. It may be noticed that the product of matrix A and matrix B is possible as the number of columns of A are equal to the number of rows of B. The product BA is not possible as the number of columns of B are not equal to rows of A.

Product	Dimensions of the Matrices	Does a product exist? (Is it possible to multiply the given matrices in this order?)	Dimensions of Product Matrix
AB	$A = 3 \times 3$ $B = 3 \times 2$ $\uparrow$ inner dimensions	Yes, the product exists since the inner dimensions match (# of columns of A = # of rows of B).	3×2
BA	$B = 3 \times 2$ $A = 3 \times 3$ $\uparrow$ inner dimensions	No, the product does not exist since the inner dimensions do not match (# of columns of B $\neq$ # of rows of A).	n/a

MULTIPLICATIVE INVERSES

Real Numbers

Two non-zero real numbers are multiplicative inverses of each other if their products, in both orders, is 1. Thus,

The multiplicative inverse of a real number, x is $\frac{1}{x}$ or x^{-1} since $x \cdot \frac{1}{x} = 1$ and $\frac{1}{x} \cdot x = 1$.

Example:

The multiplicative inverse of 5 is $\frac{1}{5}$ since

$$5 \cdot \frac{1}{5} = 1 \text{ and } \frac{1}{5} \cdot 5 = 1$$

Matrices

16-Matrices

Two 2×2 matrices are inverses of each other if their products, in both orders, are the 2×2 identity matrix. Thus, the multiplicative inverse of a 2×2 matrix, A is A^{-1} since $A^{-1} = I_2$ and $A^{-1} \cdot A = I_2$.

Example:

The multiplicative inverse of a matrix, $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ is $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.

$$\left. \begin{aligned} \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned} \right\} \end{matrix}$$

LECTURE 11
MATRICES**OBJECTIVES**

The objectives of the lecture are to learn about

- Review Lecture 10
- Matrix functions in Excel
- Set up and manipulate ratios.
- Allocate an amount on a prorata basis using proportions.

MATRIX FUNCTIONS IN MS EXCEL

The Matrix Functions in Microsoft Excel are as follows:

1. MINVERSE
2. MDETERM
3. MMULT

MINVERSE

Returns the inverse matrix for the matrix stored in an array

Syntax

MINVERSE(array)

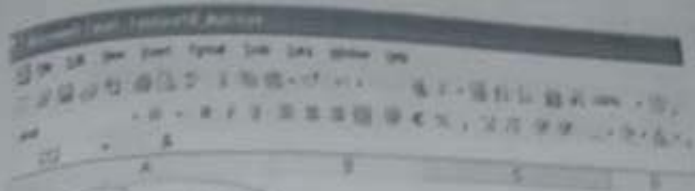
array is a numeric array with an equal number of rows and columns

Remarks

- Array can be given as a cell range, such as A1:C3; as an array constant, such as {1,2,3,4,5,6,7,8,9}; or as a name for either of these.
- If any cells in array are empty or contain text, MINVERSE returns the #VALUE! error value.
- MINVERSE also returns the #VALUE! error value if array does not have an equal number of rows and columns.
- Formulas that return arrays must be entered as array formulas.
- Inverse matrices, like determinants, are generally used for solving systems of mathematical equations involving several variables. The product of a matrix and its inverse is the identity matrix — the square array in which the diagonal values equal 1, and all other values equal 0.
- As an example of how a two-row, two-column matrix is calculated, suppose that the range A1:B2 contains the letters a, b, c, and d that represent any four numbers. The following table shows the inverse of the matrix A1:B2.

	Column A	Column B
Row 1	$d/(a*d-b*c)$	$b/(b*c-a*d)$
Row 2	$c/(b*c-a*d)$	$a/(a*d-b*c)$

- MINVERSE is calculated with an accuracy of approximately 16 digits, which may lead to a small numeric error when the cancellation is not complete.
- Some square matrices cannot be inverted and will return the #NUM! error value with MINVERSE. The determinant for a noninvertible matrix is 0.



MINVERSE

Data Data

4 -1

2 0

0 0.5

-1 2

(=MINVERSE(A4:B5))

MDETERM

Returns the matrix determinant of an array.

Syntax

MDETERM(array)

array is a numeric array with an equal number of rows and columns.

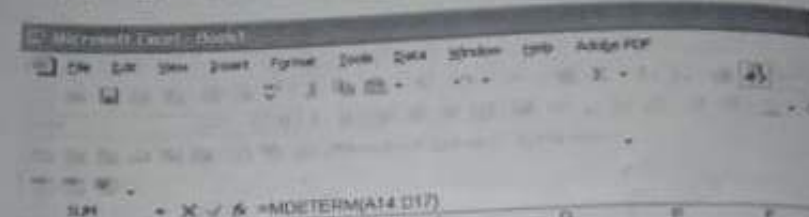
Remarks

- Array can be given as a cell range, for example, A1:C3, as an array constant, such as {1,2,3,4,5,6,7,8,9}, or as a name to either of these.
- If any cells in array are empty or contain text, MDETERM returns the #VALUE! error value.
- MDETERM also returns #VALUE! if array does not have an equal number of rows and columns.
- The matrix determinant is a number derived from the values in array. For a three-row, three-column array, A1:C3, the determinant is defined as:

$$\text{MDETERM}(A1:C3) = A1*(B2*C3-B3*C2) + A2*(B3*C1-B1*C3) + A3*(B1*C2-B2*C1)$$
- Matrix determinants are generally used for solving systems of mathematical equations that involve several variables.
- MDETERM is calculated with an accuracy of approximately 16 digits, which may lead to a small numeric error when the calculation is not complete. For example, the determinant of a singular matrix may differ from zero by 1E-16.

MDETERM-EXAMPLE

The example shows an array of dimension 4 x 4 in cell range A14:D17. The formula was entered in cell A18. The result of this calculation is 88.



The screenshot shows the Microsoft Excel interface. The formula bar displays `=MDETERM(A14:D17)`. The worksheet grid shows the following data:

	A	B	C	D	E	F
11	MDETERM					
12						
13		Data	Data	Data	Data	
14		✓ 1	3	8	5	
15		1	3	6	1	
16		1	1	1	0	
17		7	3	10	✓ 2	
18	=MDETERM(A14:D17)					
19	Determinant of the matrix above (88)					

There are other ways also for using this function. For example you can enter the matrix as an array constant

`=MDETERM((3,6,1,1,1,0,3,10,2))`

Determinant of the matrix as an array constant (1)

You can calculate the determinant of the matrix in the array constant.

`=MDETERM((3,6,1,1))`

Determinant of the matrix in the array constant (-3)

Unequal number of rows and columns results in an error.

`=MDETERM((1,3,8,5,1,3,6,1))`

Returns an error because the array does not have an equal number of rows and columns (#VALUE!)

MMULT

Returns the matrix product of two arrays. The result is an array with the same number of rows as array1 and the same number of columns as array2.

Syntax

MDTERM-EXAMPLE

The example shows an array of dimension 4 x 4 in cell range A14:D17. The formula was entered in cell A18. The result of this calculation is 88.

The screenshot shows an Excel spreadsheet with the following content:

- Formula Bar:** Displays the formula `=XLOOKUP(A4:D17)`.
- Worksheet Grid:**
 - Row 13: Column headers "Data", "Data", "Data", "Data".
 - Row 14: Values 1, 3, 8, 5. A checkmark is in the first cell.
 - Row 15: Values 1, 3, 6, 1.
 - Row 16: Values 1, 1, 1, 0.
 - Row 17: Values 7, 3, 10, 2. A checkmark is in the last cell.
 - Row 18: Cell A18 contains the formula `=MDETERM(A14:D17)`.

The matrix being calculated is:

Data	Data	Data	Data
1	3	8	5
1	3	6	1
1	1	1	0
7	3	10	2

There are other ways also for using this function. For example you can enter the matrix as an array constant.

```
=MDETERM(3.6,1.1,1.0,3.10,2))
```

Determinant of the matrix as an array constant (t)

You can calculate the determinant of the matrix in the array constant

```

=MDTERM(3,6,1,1))

```

Determinant of the matrix in the array
constant (-3)

Unequal number of rows and columns results in an error.

=MDETERM({1,3,8,5,1,3,5,1}))

Returns an error because the array does not have an equal number of rows and columns (#VALUE!).

MULT

MMULT
Returns the matrix product of two arrays. The result is an array with the same number of rows as array1 and the same number of columns as array2.
Syntax
MMULT(array1,array2)
Array1: array of numbers or references to a range of numbers.

Syntax

MMULT(array1,array2)
Array1: array 2

Remarks

- The number of

Remarks

- The number of columns in array1 must be the same as the number of rows in array2, and both arrays must contain only numbers.
- Array1 and array2 can be given as cell ranges, array constants, or references.

- If any cells are empty or contain text, or if the number of columns in array1 is different from the number of rows in array2, MMULT returns the #VALUE! error value.
- The matrix product array a of two arrays b and c is

$$a_{ij} = \sum_{k=1}^n b_{ik} c_{kj}$$

where i is the row number, and j is the column number.

Formulas that return arrays must be entered as array formulas.

MMULT-EXAMPLE

Let

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}. \text{ Find } AB$$

To find the product AB, follow these steps.

- Enter array1 in cell range A25:B26, and array2 in cell range D25:E26.

Microsoft Excel - Book1						
File Edit View Insert Format Tools Data Window Help						
Arial 10 B I U						
D37 %						
	A	B	C	D	E	F
23	MMULT					
24	array1			array2		
25	1	3		2	0	
26	7	2		0	2	
27						
28	Product of array1 and array2					
29	2	6	=MMULT(A25:B26,D25:E26)			
30	14	4				
31						

- Find the dimension of AB matrix. Here as A and B are 2×2 matrices so AB is also a 2×2 matrix.
- Click on cell A29.
- Keeping left mouse button pressed, drag it to cell B30. Four cells A29, B29, A30, B30 will be selected.
- Press F2 from your keyboard. (Key F2 is selected to enter Edit Mode in the active cell. It's a keyboard shortcut. Even if you don't press F2, you can write the formula).
- Type the formula =MMULT(
- Select array1.
- Put comma.
- Select array2.
- Close bracket
- Press Ctrl, Shift, Enter keys simultaneously from your keyboard.
- Product AB will appear in cells A29, A30, B29, B30.

RATIO

A Ratio is a comparison between things. If in a room there are 30 men and 15 women then the ratio of men to women is 2 to 1. This is written as 2:1 and read as "two is to one". ":" is the notation for a ratio. Be careful, order matters. A ratio of 2:1 is not the same as 1:2. In the form of fraction, we can write 2:1 as $\frac{2}{1}$.

The method of calculating ratios is as under:

1. Find the minimum value
2. Divide all the values by the smallest value.

In the above example, the smallest value was 15. Division gives $\frac{30}{15} = 2$ for men and $\frac{15}{15} = 1$ for women. The ratio is therefore 2:1 for men and women.

RATIO-EXAMPLE

Three friends Ali, Fawad and Tanveer are doing business together. To set up the business Ali invested Rs. 7800, Fawad Rs. 5,200 and Tanveer Rs. 6,500. What is the ratio of their investments. As discussed above the smallest value is 5200Rs. All values are divided by 5200. The results are 1.5 for Ali, 1 for Fawad and 1.25 for Tanveer. The answer is: 1.5 : 1 : 1.25.

Microsoft Excel - Lecture10, Ratio, Proportion

	A	B	C	D
52				
53				
54				
55				
56				
57	Ali	7800		1.5
58	Fawad	5200		1
59	Tanveer	6500		$=B59/B58$
60				1.25
61				

This example can be solved in Excel. The formula is as under:

Cell D57: $=B57 / B58$

Cell D58: $=B58 / B58$

Cell D59: $=B59 / B58$

The result for cell D59 is shown in cell D60, because the cell D59 is used to display the formula.

PROPORTION

A proportion is an equation with a ratio on each side. It is a statement that two ratios are equal.

$3:4 = 6:8$ OR $3/4 = 6/8$ is an example of a proportion.

When one of the four numbers in a proportion is unknown, cross products may be used to find the unknown number. This is called solving the proportion or **ESTIMATING USING RATIO**.

EXAMPLE

Ratio of sales of Product X to sales of Product Y is 4:3. The sales of product X is forecasted at Rs. 180,000. What should be the Sales of product Y to maintain the ratio of sales between the two products.

CALCULATION

Ratio sales X : Y = 4 : 3

Insert the value for forecasted sale for X.

$180,000 : Y = 4 : 3$

It can be rewritten as:

$180,000/Y = 4/3$

Cross - multiply

$180,000 \times 3 = 4 \times Y$

Rewrite to bring the unknown to the left of the equality

$4 \times Y = 180,000 \times 3$

Solve

$Y = (180,000 \times 3)/4$

$Y = 135,000$ Rs

Calculations using EXCEL

In cells B70 and B71 the ratios of Product X and Y were entered.

The value of forecast of product X was entered in cell D70.

Before writing down the formula in excel, it was derived as follows:

1. Ratio of X = (cell B70)

2. Ratio of y = (cell B71)

3. Sale of X = (cell D70)

4. Sale of Y = (cell D71)

Now Ratio X : Y = (cell B70) / (cell B71)

Ratio of sales = (cell D70) / (cell D71)

Cross-multiply

$(\text{cell B70}) \times (\text{cell D71}) = (\text{cell B71}) \times (\text{cell D70})$

Cell D71 is unknown. Hence:

$(\text{cell D71}) = (\text{cell B71}) \times (\text{cell D70}) / (\text{cell B70})$

Or

$(\text{cell D71}) = (\text{cell B71}) / (\text{cell B70}) * (\text{cell D70})$

Thus the formula was:

$=B71/B70*D70$

Please note that actually we are using the ratio Y to X as it is easier to think of ratio of unknown to the known.

in a 500 bed hospital there are 200 nurses and 150 other staff. If the hospital extends by a new wing for 100 beds, then what additional staff is needed? Let
 500 beds B_1 and 100 beds B_2 . Staff nurses N_1 is 200 and other staff O_1 is 150. What is the value of N_2 and O_2 for B_2 ? Obviously the ratio of beds will be used. As pointed out above, think of the ratio of unknown to known. In other words ratio B_2/B_1 or B_2/B_1 . Ratio of nurses would be N_2/N_1 . Ratio of other staff would be O_2/O_1 .
 Now
 $N_2/N_1 = B_2/B_1$ Or $N_2 = (B_2/B_1) * N_1$ or $N_2 = (100/500) * 200 = 40$ Nurses
 $O_2/O_1 = B_2/B_1$ Or $O_2 = (B_2/B_1) * O_1$ or $O_2 = (100/500) * 150 = 30$ other staff

Calculation

Beds	Nurses	Other staff
500	200	150
100	X?	Y?

Nurses

$$500 : 200 = 100 : X$$

$$500 X = 200 \times 100$$

$$X = (200 \times 100) / 500 = 40$$

Other staff

$$Y = (150 \times 100) / 500 = 30$$

Calculation using EXCEL

The calculation using EXCEL was done in a similar fashion as the previous example. The calculation is self-explanatory.

Microsoft Excel - Lecture10 Ratios Proportions

File Edit View Insert Format Tools Data Window Help Adobe PDF

Σ

SUM X ✓ f_x =(D7/D5*E5)

A B C D E F G H I

USING RATIOS

Hospital Addition

5	Beds	500	100	
6	Nurses	200	?	40
7	Other staff	150	?	=(D7/D5*E5)
8				30

Calculation

Mango juice : Apple juice : Orange juice
 3 : 2 : 1

Total = 6

X?

Y?

Z?

Total = 2 litre

Mango juice (X) = $(3/6) \times 2 = 1$ litre

Apple juice (Y) = $(2/6) \times 2 = 0.67$ litre

Orange juice (Z) = $(1/6) \times 2 = 0.33$ liter

Calculation using EXCEL

Here also the similar ratios were used.

Mango = $B20/B23 \times D23$

Apple = $B21/B23 \times D23$

Orange = $B22/B23 \times D23$

Microsoft Excel - Lecture10 Ratios Proportions

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Σ

SUM X / % =B22/B23*D23

	A	B	C	D	E	F	G
15							
17	USING RATIOS						
18							
19		RATIO					
20	Mango juice	3		?	1		
21	Apple juice	2		?	0.7		
22	Orange juice	1		$=B22/B23 \times D23$			
23	Total. Litre	6					
24				2			

- Module 3 has the following content:
- Ratio and Proportions
 - Merchandising (Lectures 12)
 - Mathematics of Merchandising (Lectures 13-16)

ESTIMATING USING RATIOS-EXAMPLE 1

In the previous lecture, we studied how ratios can be used to determine unknowns. Here is another example with a slightly different approach. Here, ratios between the quantities and data of only one quantity is known. We will estimate the total quantity that can be made? It is the quantity of orange juice that will determine the total quantity that can be made. Again the method is to use the ratio of the unknown to the known.

Punch recipe

In a punch the ratio of mango juice, apple juice and orange juice is 3:2:1. If you have 1.5 litres of orange juice, how much punch can you make?

Calculation

Mango juice : Apple juice : Orange juice
 $\frac{3}{3} : \frac{2}{2} : \frac{1}{1}$

$$\text{Total} = 3+2+1 = 6$$

X (say) : Y (say) : Z (say)
 $\frac{?}{?} : \frac{?}{?} : \frac{1.5 \text{ litres}}{1}$

$$\text{Total} = ? \text{ litre}$$

$$\text{Mango juice (X)} = (3/1) \times 1.5 = 4.5 \text{ litre}$$

$$\text{Apple juice (Y)} = (2/1) \times 1.5 = 3.0 \text{ litre}$$

$$\text{Orange juice (Z)} = 1.5 \text{ litre}$$

$$\text{Punch} = 4.5 + 3.0 + 1.5 = 9 \text{ litres}$$

EXCEL Calculation

The method used is the same as in previous examples.

USING RATIOS

	RATIO		
Mango juice	3	?	4.5
Apple juice	2	?	3
Orange juice	1	1.5	1.5
Total. Litre	6	$=B34/B33*D33$	
		9	

ESTIMATING USING RATIOS-EXAMPLE 2

In a punch ratio of mango juice, apple juice and orange juice is 3 : 2 : 1.5. If you have 500 litres of orange juice, find how much mango and apple juices are required to make the punch.

Punch recipe

The ratio of mango juice, apple juice and orange juice is 3 : 2 : 1.5. If you have 500 milliliters of orange juice, how much mango juice and apple juice is needed?

Mango juice : Apple juice : Orange juice

3 : 2 : 1.5

Total = 6.5

X (say) : Y (say) : Z (say)

? : ? : 500 litres

Total = ? litres

Mango juice (X) = $(3/1.5) * 500 = 1000$ litre

Apple juice (Y) = $(2/1.5) * 500 = 667$ litre

Orange juice (Z) = 500 litre

Punch = $1000 + 667 + 500 = 2167$ litre

EXCEL Calculation

Here also ratios were used.

Mango juice = $B45/B47*D47$

Apple juice = $B46/B47*D47$

Orange juice = D47

USING RATIOS

	RATIO	
Mango juice	3	
Apple juice	2	
Orange juice	1.5	
Total. MI	6.5	
		$\frac{1000}{=B45/B47*D47}$
		?
		667
		500
		?
		2167

EXAMPLE

In a certain class, the ratio of passing grades to failing grades is 7 to 5. How many of the 36 students failed the course?
 The ratio, "7 to 5" (or $7:5$ or $7/5$), tells you that, of every $7 + 5 = 12$ students, five failed.
 That is, $5/12$ of the class failed.
 Then $(5/12)(36) = 15$ students failed.

PROPORTION

$$a/b = c/d$$

the values in the "b" and "c" positions are called the "means" of the proportion, while the values in the "a" and "d" positions are called the "extremes" of the proportion. A basic defining property of a proportion is that the product of the means is equal to the product of the extremes. In other words, given

$$a/b = c/d$$

it is a fact that $ad = bc$.

PROPORTION-EXAMPLES

Is $24/140$ proportional to $30/176$?

Check:

$$140 \times 30 = 4200$$

$$24 \times 176 = 4224$$

So the answer is that given ratios are not proportional.

PROPORTION EXAMPLE 1

Find the unknown value in the proportion: $2 : x = 3 : 9$.

$$2 : x = 3 : 9$$

First, convert the colon-notation ratios to fractions:

$$2/x = 3/9$$

Cross multiply

$$18 = 3x$$

$$6 = x$$

PROPORTION EXAMPLE 2Find the unknown value in the proportion: $(2x + 1) : 2 = (x + 2) : 5$

$$(2x + 1) : 2 = (x + 2) : 5$$

First, convert the colon-notation ratios to fractions:

$$(2x + 1)/2 = (x + 2)/5$$

Then solve:

$$5(2x + 1) = 2(x + 2)$$

$$10x + 5 = 2x + 4$$

$$8x = -1$$

$$x = -1/8$$

MERCHANDISING

What does merchandising cover?

- Understand the ordinary dating notation for the terms of payment of an invoice.
- Solve merchandise pricing problems involving mark ups and markdowns.
- Calculate the net price of an item after single or multiple trade discounts.
- Calculate a single discount rate that is equivalent to a series of multiple discounts.
- Calculate the amount of the cash discount for which a payment qualifies.

STAKEHOLDERS IN Merchandising

Who are the stakeholders in merchandising?

The main players are:

- **Manufacturer**
- **Middleman**
- **Retailer**
- **Consumer**

There are discounts at all levels in the above chain.

MIDDLEMAN

A middle man is a person who buys a product directly from the manufacturer, and then either sells the product at retail prices to the public, or sells the product at wholesale prices to a distributor. There can often be more than one middle man when the latter practice is adopted. A middle man can purchase from the manufacturer and then work with another middle man who buys for the distributor. The manufacturer often views the middle man as the alternative to direct distribution.

List price or Retail price

List price refers to the manufacturer's suggested retail pricing. It may or may not be the price asked of the consumer. Much depends on

1. the product itself,
2. the built-in profit margin,
3. Supply and demand

A product that is in high demand with low availability will sometimes sell higher than the list price though this is less common than the reverse.

Virtually all products have a suggested retail or list price. Resellers (middleman, retailer) buy products in bulk and get a substantial discount in order to be able to get profit from selling the product at or below list price.

Trade Discount

Let L is the list price, then amount of trade discount is some percentage of this price. List price less amount of discount is the net price. In mathematical terms, we can write:

$$\text{Amount of discount} = d \times L$$

Where, d = Percentage of Discount

L = List Price

$$\text{Net Price} = L - Ld = L(1 - d)$$

Net Price = List Price - Amount of Discount

LECTURE 13 MATHEMATICS OF MERCHANDISING

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 12
- Solve merchandising pricing problems involving markup and markdown

MARKUP

Markup is an amount added to a cost price while calculating a selling price. Especially, an amount that takes into account of overhead and profit. Markup can be expressed as:

1. Percentage of cost.
2. Percentage of sale.
3. Rs Markup

Markup as Percentage of Cost (MUC):

Here markup is some percentage of cost price. For simplicity, it is also named as %Markup on cost. The relation between %markup on cost, cost price and selling price is:

$$\begin{aligned}\text{Selling Price} &= \text{Cost price} + (\text{Cost price} \times \% \text{Markup on cost}) \\ &= \text{Cost price} (1 + \% \text{Markup on cost})\end{aligned}$$

Markup as Percentage of Sale price (MUS):

Here markup is some percentage of selling price. For simplicity, it is also named as %Markup on sale. The relation between %markup on sale, cost price and selling price is:

$$\begin{aligned}\text{Selling Price} &= \text{Cost price} + (\text{Selling price} \times \% \text{Markup on sale}) \\ \text{Cost price} &= \text{Selling price} - (\text{Selling price} \times \% \text{Markup on sale}) \\ &= \text{Selling price} (1 - \% \text{Markup on sale})\end{aligned}$$

Rs Markup:

Markup in terms of rupees is called Rs markup. The relations between Rs markup, cost price and selling price are:

1. Selling Price = Cost price + Rs Markup
2. Rs Markup = %Markup on cost $\times$ Cost price
3. Rs Markup = %Markup on sale $\times$ Selling price

Any of the above formula can be used to find Rs Markup.

For example:

The cost price of certain item is 80Rs and its selling price is 100Rs. Then

$$\begin{aligned}\text{Rs Markup} &= \text{Selling price} - \text{Cost price} \\ &= 100 - 80 \\ &= 20 \text{ Rs}\end{aligned}$$

Remember: If some percentage is given as markup, without mentioning that whether it is markup on cost or markup on sale, it is evident that %markup on cost is under consideration.

EXAMPLE 1

A golf shop pays its wholesaler 2,400Rs. for a certain club, and then sells it for 4,500Rs. What is the markup rate?

Calculation of Markup

Cost price = 2400Rs

Selling price = 4500Rs

$$\text{Selling Price} = \text{Cost price} + (\text{Cost price} \times \% \text{Markup on cost})$$

$$\% \text{Markup on cost} = \frac{\text{Selling price} - \text{Cost price}}{\text{Cost price}} \times 100\%$$

Since

$$\text{Rs Markup} = \text{Selling Price} - \text{Cost price}$$

$$\% \text{Markup on cost} = \frac{\text{Rs markup}}{\text{Cost price}} \times 100\%$$

First, calculate Rs markup
 Rs markup = 4500 - 2400 = 2100Rs
 Then %markup on cost
 $\% \text{Markup on cost} = \frac{\text{Rs markup}}{\text{Cost price}} \times 100\%$

$$= 2100/2400 \times 100\% = 87.5\%$$

Remember to convert this fraction value to a percent. The markup rate is 87.5%.

Calculation using EXCEL

Enter whole-sale price 2400 in cell B5

Enter sale price 4500 in cell B6

Enter formula for Rs. Markup =B6-B5 in cell B7 and press enter. The answer is 2100.

Enter formula for % markup =B7/B5*100 in cell B8 and press Enter. The answer is 87.5% shown in cell B9.

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Standard toolbar icons: Undo, Redo, Cut, Copy, Paste, Find, Replace, etc.

Formatting toolbar icons: Bold, Italic, Underline, Font color, Background color, etc.

Formulas toolbar icons: Sum, Average, etc.

Formula bar: SUM X ✓ f_x =B7/B5*100

	A	B	C	D
1				
2	MARKUP			
3				
4				
5	Whole-sale price	2400		
6	Sale price	4500		
7	Markup Rs.	2100		
8	% Markup	=B7/B5*100		
9		87.5		
10				

Enter whole-sale price 1500 in cell B17.
 Enter % Markup in cell B18.
 Enter formula $=(1+B18/100)*B17$ in cell B19. Here the term $1+B18/100$ is the **multiplication factor**. B18/100 is the %markup in fraction. The answer 2100 is shown in cell B20.
 We could have calculated the multiplication factor separately. But as you see it is not necessary as the entire calculation can be done in one line.

Microsoft Excel - Book1

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Formula Bar: $= (1 + B18/100) * B17$

	A	B	C	D
12				
13				
14	MARKUP			
15	Example 2			
16				
17	Whole-sale price	1500		
18	% Markup	40		
19		$=(1+B18/100)*B17$		
20		2100		
21				

MARKDOWN

* Markdown means a reduction from the original sale price to

1. stimulate demand or
2. take advantage of reduced costs or
3. force competitors out of the market.

* Markdown can be expressed as

- ❖ Percentage of current selling price
- ❖ Rs markdown

Markdown as Percentage of current selling price

Here markdown is some percentage of current selling price. For simplicity, it is also named as percent markdown (%markdown). The relation between current selling price, %markdown and new selling price is:

$$\begin{aligned}\text{New selling price} &= \text{Current selling price} - (\text{Current selling price} \times \% \text{ markdown}) \\ &= \text{Current selling price} (1 - \% \text{ markdown})\end{aligned}$$

Rs Markdown

Markdown in terms of rupees is called Rs markdown.

$$\begin{aligned}1. \text{ New selling price} &= \text{Current selling price} - \text{Rs Markdown} \\ 2. \text{ Rs markdown} &= \text{Current selling price} \times \% \text{ markdown}\end{aligned}$$

Let us look at an example to understand how markdown is calculated.

MARKDOWN-EXAMPLE 1

An item originally priced at 3,300 Rs. is marked 25% off.
What is the sale price?

Markdown

First, find the Rs markdown.

The markdown is 25% of the original price or current selling price, as
Rs markdown = original price $\times$ % markdown

$$= (0.25)(3300) = 825 \text{Rs}$$

Selling Price

Then calculate the sale price, by subtracting the markdown from the original price:

$$\text{New Selling price} = 3,300 - 825 = 2,475 \text{Rs}$$

The sale price is 2,475 Rs.

Calculation using EXCEL

Enter original price 3300 in cell B28.

Enter % Markdown 25 in cell B29.

Enter formula for Rs. Markdown $(=B29/100*B28)$ in cell B30. Here the term $B29/100$ is the markdown in fraction. The result of this part of the calculation is 825.

Enter formula for new sale price $(=B28-B30)$ in cell B31. This formula is not shown in the slide.

We could have calculated the new sale price directly also by writing just one formula $(=1-B29/100)*B28)$ by using following formula

$$\text{New selling price} = \text{Current selling price} (1 - \% \text{ markdown})$$

In other words the multiplication factor is calculated as $1 - 0.25 = 0.75$ and multiplied with the original price 3300. The answer would be the same. By breaking the calculation in parts you can check the intermediate result and avoid errors. But if you become very conversant with formulas then you may wish to reduce the number of unnecessary steps in the calculations.

MARKDOWN

Example 1

Original price

3300

% Markdown

25

$$=B29/100*B28$$

Sale price

2475

DISCOUNT

Discount is a reduction in price which the seller offers to the buyer. There are different types of discount.

1. Trade discount
2. Cash discount
3. Seasonal discount, etc

TRADE DISCOUNT

When a manufacturer or wholesaler offers goods for sale, a list price or retail price is set for each item. This is the suggestion price to be charged from the ultimate consumer. A discount on the list price granted by a manufacturer or wholesaler to buyers in the same trade is called trade discount. Trade discount represents a reduction in list price in return for quantity purchases. Thus

$$\begin{aligned} \text{Rs Trade discount} &= \text{list price} \times \text{discount rate} \\ \text{Net price} &= \text{list price} - \text{Rs trade discount} \end{aligned}$$

There are two main types of trade discounts

1. Single trade discount
2. Multiple or series trade discount

SINGLE TRADE DISCOUNT-EXAMPLE 1

The price of office equipment is 3000 Rs. The manufacturer offers a 30% trade discount. Find the net price and the trade discount amount.

Discount

$$\begin{aligned} \text{Net Price} &= L(1 - d) \\ &= 3000(1 - 0.3) \\ &= 3000(0.7) \\ &= 2100 \text{ Rs} \end{aligned}$$

$$\begin{aligned} \text{Amount of discount} &= dL \\ &= 0.3 \times 3000 = 900 \text{ Rs} \end{aligned}$$

Calculation using EXCEL

Enter price of equipment 3000 in cell B39

... refers to the giving of further discounts... such discount is offered for selling product in bulk.

If series discount of 15%, 10%, 5% are offered on list price, say L , of an item then net price is calculated as follows-

Subtract 15% of L from L . Let the new price is L_1 .

$$L_1 = L - (L \times 15\%)$$

Then subtract 10% of L_1 from L_1 . Let the new price is L_2 .

$$L_2 = L_1 - (L_1 \times 10\%)$$

Then subtract 5% of L_2 from L_2 . The new price is net price of an item.

$$N = L_2 - (L_2 \times 5\%)$$

Or alternatively

$$N = L (1 - 15\%) (1 - 10\%) (1 - 5\%)$$

Let $d_1 = 15\%$, $d_2 = 10\%$, $d_3 = 5\%$, then above formula becomes

$$N = L (1 - d_1)(1 - d_2)(1 - d_3)$$

Remember: total discount is not $15\% + 10\% + 5\% = 30\%$.

SERIES TRADE DISCOUNT-EXAMPLE

The price of office furniture is Rs. 20,000

The series discounts are:

20%, 10%, 5%

What is the net price?

For series trade Discount

$$\text{Net price} = (1 - d_1)(1 - d_2)(1 - d_3)$$

Here $d_1 = 20\%$, $d_2 = 10\%$, $d_3 = 5\%$. So

$$\begin{aligned}\text{Net price} &= 20,000(1 - 0.2)(1 - 0.10)(1 - 0.05) \\ &= 20,000(0.8)(0.9)(0.95) \\ &= 20,000(0.6840) \\ &= \mathbf{13,680 \text{ Rs.}}\end{aligned}$$

$$\text{net price} = L (1 - d)$$

TRADE DISCOUNT

Gross price Rs.

First discount %

Next discount % on first

Next discount % on first

Net price. Rs. (13680)

20000
20
10
5

$$=B77*(1-B78/100)*(1-B79/100)*(1-B80/100)$$

LIST PRICE

An order for power tools has a Rs. 2100 net price after a 30% trade discount. is the list price?

Net Price

$$\text{Net Price} = L(1 - d)$$

$$L = N / (1 - d)$$

$$= 2100 / (1 - 0.3)$$

$$= 2100 / (0.7)$$

$$= 3000 \text{ Rs.}$$

EXCEL Calculation

EXCEL formula for list price was based on the calculation $= 2100 / (1 - 0.3)$.

The net price was entered in cell B67.

% Trade discount was entered in B69.

The formula for list price was entered in cell B71 as $=B67 / (1-B69/100)$.

The answer is shown in cell C72 as 3000.

DISCOUNT PERIODS

Discount Periods are periods for the buyer to take advantage of Discount Terms.

CREDIT PERIODS

Credit Periods are periods for the buyers to pay invoices within specified times.

CASH DISCOUNT-EXAMPLE

Invoice was dated May 1st. The terms 2/10 mean that 2% discount is offered if invoice is paid up to 10th May.
What is the net payment for invoice value of Rs. 50,000 if paid up to 10th May?

Cash Discount

$$\begin{aligned} N &= L(1 - d) \\ &= 50,000(1 - 0.02) \\ &= 50,000(0.98) \\ &= 49,000 \text{ Rs.} \end{aligned}$$

EXCEL Calculation

EXCEL formula for net price was based on the calculation $= 50000 * (1 - 0.02)$.
However, here an IF condition was applied, that means that if the payment date in cell D31 (\$ sign is put in front of row and column to fix its location) is less than or equal to 10 May then the discount will be as given in cell D30. Here also \$ sign was used to fix the location of the cell.
In cell D38, the date was changed to 11 May and the same formula was applied again. The result as shown in cell D39 and D40 are 0% (for % discount) and 0 Rs (for Rs. Discount).

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D45							
	A	B	C	D	E	F	G
29	CASH DISCOUNT						
30	Discount			2 %			
31	Last date			10 May			
32							
33	Invoice value			50000 Rs			
34							
35	Payment date			9 May			
36	Discount			=IF(\$D\$35<=\$D\$31,\$D\$30,0)			
37				1000 Rs			
38	Payment date			11 May			
39	Discount			0 %			
40				0 Rs			

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s or 100

LECTURE 15 MATHEMATICS OF MERCHANDISING PART 3

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 14
- Financial Mathematics Part 2

PARTIAL PAYMENTS

When you buy on credit and have cash discount terms, part of the invoice within the specified time. These part payments are called **Partial Payments**.

Let us look at an example:

You owe Rs. 40,000.

Your terms were 3/10 (3% discount by 10th day).

Within 10 days you sent in a payment of Rs. 10,000.

Rs. 10,000 was a part payment.

How much is your new balance?

First we will find the amount that if 3% discount is given on it, the net amount is 10,000Rs.

Let that amount is t . Then

$$10000 = t(1 - 0.03)$$

$$\text{This implies, } t = \frac{10000}{(1 - 0.03)}$$

$$\text{Thus } t = 10309\text{Rs}$$

This means that although you pay 10,000Rs, due to 3% cash discount among 40,000Rs is paid.

$$\text{Hence the new balance} = 40000 - 10309 = 29691\text{Rs.}$$

MARKETING TERMS

There are a number of marketing terms.

First of these is the **Manufacturer Cost**. This is the cost of manufacturing.

Next is the price charged to middlemen in 'The Distribution Chain'.

The Distributor > Wholesaler > Retailer is a chain.

The next term is the **Selling Price**. This is the price charged to Consumers by Retailers. It may or may not be the same as list price.

MARKETING, OPERATING EXPENSES AND SELLING PRICE

Gross Sales less Cost of Goods sold gives the Gross Profit. Then less the Operating Expenses gives the Net Profit.

Operating Expenses

(without discount)
→ amount
Partial payment = t (1 - discount %)
↓
Discount

LECTURE 15
MATHEMATICS OF MERCHANDISING
PART 3

OBJECTIVES

- The objectives of the lecture are to learn about
- Review Lecture 14
 - Financial Mathematics Part 2

PARTIAL PAYMENTS

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Let that amount be t . Then
 $10000 = t(1 - 0.03)$

This implies, $t = \frac{10000}{(1 - 0.03)}$

Thus, $t = 10309$ Rs

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Hence the new balance = $40000 - 10309 = 29691$ Rs.

MARKETING TERMS

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MARKETING, OPERATING EXPENSES AND SELLING PRICE

Gross Sales less Cost of Goods sold gives the Gross Profit. The gross Profit less the Operating Expenses gives the Net Profit.

Operating Expenses

Expenses the company incurs in operating the business, e.g. rent, wages and utilities is called operating Expenses.

Selling Price

Selling Price is composed of Cost and Rs Markup.

$\text{Selling Price (S)} = \text{Cost (C)} + \text{Rs Markup (M)}$ Profit

MARGIN

While determining Sale Price, a company includes the operating expenses and profit to their own cost. This amount is called the margin of the company. It is usually calculated as percentage but can also be expressed as rupees. It is also named as **markup on sale**.

$$\text{Margin or markup on sale} = \frac{\text{Selling price} - \text{Cost price}}{\text{Selling Price}} \times 100\%$$

$$\text{Selling price} = \text{Cost price} + \text{Rs Margin}$$

Margin and markup confuse many. By margin, company evaluates that for every rupee generated in sales, how much is left over to cover basic operating costs and profit. Markup represents the amount added to a cost to arrive at a selling price.

$$\text{Markup on cost} = \frac{\text{Selling price} - \text{Cost price}}{\text{Cost price}} \times 100\%$$

For example, an item costs 50Rs, and is sold for 100Rs.

$$\text{Markup} = \frac{100 - 50}{50} \times 100\% = 100\%$$

$$\text{Margin} = \frac{100 - 50}{100} \times 100\% = 50\%$$

Note: Remember unless it is mentioned that markup is on sale, simple markup means markup on cost.

Example

A computer's cost is 9000Rs. An amount of Rs. 3,000 was added to this cost by the retailer to determine the sale price for the consumer.

Thus, the selling price

$$= 9,000 + 3,000 \text{ Rs.}$$

$$= 12,000 \text{ Rs.}$$

Rs. 3,000 is Margin available to meet Expenses and make a Profit.

MARKUP

If the Markup on cost 33% then...

$$\text{Selling Price (S)} = \text{Cost (C)} + [\text{Cost (C)} \times \text{Markup on cost (MUC)}]$$

$$S = C + (C \times \text{MUC})$$

MARKUP-EXAMPLE

You buy candles for Rs. 10.

You plan to sell them for Rs.15.

What is your Rs. Markup?

What is your percent Markup on cost?

$$\text{Rs Markup} = \text{Selling price} - \text{cost price}$$

$$\text{Selling price} - \text{Cost} = 15 - 10$$

$$= \text{Rs Markup} = \text{Rs. 5}$$

$$\text{Markup on cost} = \frac{\text{Selling price} - \text{Cost price}}{\text{Cost price}} \times 100\%$$

$$\% \text{Markup} = 5/10 \times 100\% = 50\%$$

BELLING PRICE

Fawad's Appliances bought a sewing machine for Rs. 1,500.

To make the desired profit, he needs a 60% Markup on Cost.

What is Fawad's Rs. Markup?

What is his Selling price?

Selling Price

$$\text{Rs Markup} = \text{Cost price} \times \% \text{Markup on cost}$$

$$\text{Rs. Markup} = 1,500 \times 0.6 = 900 \text{ Rs.}$$

$$\text{Selling Price (S)} = \text{Cost (C)} + \text{Rs Markup (M)}$$

$$\text{Selling Price} = 1,500 + 900 = 2,400 \text{ Rs.}$$

Or
Alternatively, since

$$\text{Selling Price (S)} = \text{Cost (C)} + \{\text{Cost (C)} \times \text{Markup on cost (MUC)}\}$$

$$S = C + (C \times \text{MUC})$$

$$= C (1 + \text{MUC})$$

$$\text{Selling price}$$

$$= 1,500 \times (1 + 0.6)$$

$$= 1,500 \times 1.6$$

$$= 2,400 \text{ Rs.}$$

EXCEL Calculation

Here 1,500 is the Sewing machine cost in cell F4 and 60 is the Percent Markup on cost in cell F5.

EXCEL formula in cell F6 for Rs. Markup was based on the calculation

$$=60/100*1500$$

The Selling price was calculated in cell F7 by using the formula =F4+F6.

The answer, 2400, is shown in cell F7.

Microsoft Excel - Book1							
File Edit View Insert Format Tools Data Window Help							
Arial 10 B I U							
F16 %							
	A	B	C	D	E	F	G
1							
2							
3							
4							
5							
6							
7							
8							

MARKUP - SELLING PRICE			
Sewing machine cost		1500	Rs
% Markup on cost		60	%
Rs Markup		=F5/100*F4	Rs
Selling price		2400	Rs

Rs. MARKUP AND PERCENT ON COST

Tarveer's flower business sells floral arrangements for Rs. 35. To make his desired profit, Tarveer needs a 40% Markup on cost. What do the flower arrangements cost Tarveer? What is the Rs. Markup?

Rs. Markup and Percent Markup on Cost

Selling price $S = \text{Cost } C + (C \times \text{Markup on cost (MUC)})$

$$S = C + 0.40(C)$$

$$35 = 1.40(C)$$

$$C = 35/1.4 = 25 \text{ Rs.}$$

$$\text{Rs. Markup} = 25 \times 0.4$$

$$= 10 \text{ Rs.}$$

EXCEL Calculation

Here 35 is the Selling price-floral arrangement in cell H15. % Markup on cost is in cell H16.

EXCEL formula in cell H18 for Cost was based on the calculation $= 35/1.4$.

The Rs. Markup was calculated in cell H19 by using the formula $= H16 \times H18/100$.

The answer, as shown in cell H19, was 10.

	A	B	C	D	E	F	G	H	I	J
15								35		
16								40%		
18								=H15/(1+H16/100)		
19								=H16*H18/100		

Rs. MARKUP AND PERCENT ON COST

Selling price-floral arrangement 35 Rs.

% Markup on cost 40 %

Cost? $= H15 / (1 + H16/100)$

Rs. Markup 10 Rs.

Formula: $S = C + M$

$$= C + 0.4C$$

$$= 1.4C$$

MARKUP / Profit

You buy candles for 2 Rs.

You plan to sell them for 2.50 Rs.

What is your Rs. Markup?

What is your Percent Markup on Selling Price?

Rs. Markup

$$\text{Rs. Markup} = 2.5 - 2 = 0.5 \text{ Rs.}$$

Percent Markup on Selling Price

As explained in lecture 13

$$\text{Cost price} = \text{Selling price} (1 - \% \text{Markup on sale})$$

$$\text{Markup on selling price} = \frac{\text{Selling price} - \text{Cost price}}{\text{Selling price}} \times 100\%$$

$$\text{Markup on Selling Price} = \frac{25.50 - 20}{25.50} \times 100\% = 21\%$$

EXCEL Calculation

Here 2 is the Purchase price in cell E30

Sale price is entered in cell E31

Rs. Markup on Purchase Price was calculated by using the formula $=E31-E30$ in cell

E32

The % Markup on sale price was calculated in cell E33 by using the formula

$$=E32/E31*100$$

The answer, as shown in cell E33, was 40.20

MARKUP AGAIN											
13	PA	SA	MR	MRP	MRP	MRP	MRP	MRP	MRP	MRP	MRP
14											
15											
16											
17											
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100											

SELLING PRICE

Fawad's Appliances bought a sewing machine for Rs. 1,500.

To make the desired profit, he needs a 60% Markup on Selling price.

What is Fawad's Rs. Markup?

What is his Selling Price?

Selling Price

As explained in lecture 13

$$\text{Selling Price} = \text{Cost price} + (\text{Selling price} \times \% \text{Markup on sale})$$

Selling Price =

$$35 = C + (0.4 \times 35)$$

$$35 = C + 14$$

$$C = 35 - 14$$

$$= 21 \text{ Rs.}$$

Or, alternatively

$$C = S - 0.4 S$$

$$= 0.6 S = 0.6 \times 35 = 21 \text{ Rs.}$$

Rs. Markup

$$\text{Rs. Markup} = 35 \times 0.4 = 14 \text{ Rs.}$$

EXCEL Calculation

Here 35 is the Sale price in cell E50.

% Markup on Sale Price is entered as 40 in cell E51.

Cost was calculated by using the formula $=E50*(1-E51/100)$. The result 21 is shown in cell D52.

EXCEL formula in cell E53 for Rs. Markup was $=E50-E52$. The result 14 is shown in cell E53.

Basic formula $S=C+0.4S$ is shown in cell A55. In cell A56 it was simplified to $0.6S=C=S(1-\text{mus})$.

Microsoft Excel - Lecture14: Markup, Discount, Markdown						
File Edit View Insert Format Tools Data Window Help						
= X ✓ / & #E50*(1-E51/100)						
	A	B	C	D	E	F
49						
50	Rs. MARKUP AND PERCENT ON COST					
51	Sale price				35 Rs.	
52	% Markup on sale price?				40 %	
53	Cost?	21	$=E50*(1-E51/100)$			
54	Rs. Markup?				14 Rs.	
55	$S=C+0.4S$					
56	$0.6S=C=S(1-\text{mus})$					
57						

CONVERTING MARKUPS

Convert 50% Markup (MU) on Cost to %MU on Sale

Formula for converting %Markup on Sale (mus) to %Markup on Cost Price (muc) is:

$$\begin{aligned} \% \text{ Markup on Selling Price (mus)} &= \\ \% \text{ Markup on Cost} / (1 + \% \text{ Markup on Cost}) \end{aligned}$$

$$\text{mus} = \text{muc} / (1 + \text{muc})$$

Solution

LECTURE 16 MATHEMATICS OF MERCHANDISING PART 4

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 15
- Markup and Markdown

Financial Mathematics Part 3

MARKDOWN

Reduction from original selling Price is called Markdown.

Formula

$$\% \text{Markdown} = (\text{Rs. Markdown} / \text{Selling Price (original)}) \times 100\%$$

MARKDOWN-EXAMPLE 1

Store A marked down a Rs. 500 shirt to Rs. 360.

What is the Rs. Markdown?

What is the %markdown?

Rs. Markdown

Let S = Sale price

$$\begin{aligned} \text{Rs. Markdown} &= \text{Old S} - \text{New S} \\ &= \text{Rs. 500} - \text{Rs. 360} \\ &= \text{Rs. 140 Markdown} \end{aligned}$$

% Markdown

$$\% \text{ Markdown} = \frac{\text{Markdown}}{\text{Old S}} \times 100\%$$

$$\begin{aligned} \% \text{ Markdown} &= \frac{140}{500} \times 100\% \\ &= 0.28 \times 100\% \\ &= 28\% \end{aligned}$$

EXCEL Calculation

Here 500 is the Original price in cell E73.

Price after Markdown is entered as 360 in cell E74.

Rs. Markdown was calculated in cell E75 by using the formula =E73-E74. The result 140 is shown in cell D75.

EXCEL formula in cell E76 for % Markdown was = E75/E73*100. The result 28 is

to cost. The applicable method should be determined first. Calculation of future forecasts can then be done through worksheets.

FORECASTS OF REVENUES

These will be done similar to the forecast of costs. Here also the method must be determined first. Once the methodology is clear, the worksheets can be prepared easily.

NET CASH FLOWS

The difference between Revenue and Cost is called the Net Cash flow. This is an important calculation as the entire Project Operation and Performance is based on its cash flows.

BENEFIT COST ANALYSIS

This is the end result of the Project Analysis. The ratio between Present Worth of benefits and Costs is called the Benefit Cost (BC) ratio. For a project to be viable without profit or loss, the BC Ratio must be 1 or more. Generally a BC Ratio of 1.2 is considered acceptable. For Public projects even lesser BC ratio may be accepted for social reasons.

INTERNAL RATE OF RETURN

Internal Rate of Return or IRR is that Discount Rate at which the Present Worth of Costs is equal to the Present Worth of Benefits. IRR is the most important parameter in Financial and Economic Analysis. There are a number of functions in EXCEL for calculation of IRR.

BREAK-EVEN ANALYSIS

In every project where investment is made it is important to know how long it takes to recover the investment. It is also important to find the breakeven point where the Cash Inflow becomes equal to Cash Outflow. After that point the company has a positive cash flow (i.e. there is surplus cash after meeting expenses).

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 16
- Financial Mathematics
- Introduction to Linear Equations

Module 4

Module 4 covers the following:

- Financial Mathematics (Lecture 17)
- Applications of Linear Equations (Lecture 17-18)
- Break-even Analysis (Lectures 19-22)
- Mid-Term Examination

PROJECT FINANCIAL ANALYSIS

Project Financial Analysis covers the following:

- Cost estimates
- Revenue estimates
- Forecasts of costs
- Forecasts of revenues
- Net cash flows
- Benefit cost analysis
- Internal Rate of Return
- Break-Even Analysis

EXCEL FUNCTIONS FOR FINANCIAL ANALYSIS

List of Excel Financial functions is as under. The name and utility of each function is given below:

AMORDEGRC

Returns the depreciation for each accounting period. If an asset is purchased in the middle of the accounting period, the prorated depreciation is taken into account. The function is similar to AMORLINC, except that a depreciation coefficient is applied in the calculation depending on the life of the assets.

Syntax

AMORDEGRC(cost,date_purchased,first_period,salvage,period,rate,basis)

Important Dates should be entered by using the DATE function, or as results of other formulas or functions. For example, use DATE(2008,5,23) for the 23rd day of May, 2008. Problems can occur if dates are entered as text.

Cost cost of the asset.

Date_purchased date of the purchase of the asset.

First_period date of the end of the first period.

Salvage salvage value at the end of the life of the asset.

Period period.

Rate rate of depreciation.

Basis year basis to be used.

365 days in a year
360 days in a year (European method)

Remarks

- Microsoft Excel stores dates as sequential serial numbers so they can be used in calculations. By default, January 1, 1900 is serial number 1, and January 1, 2008 is serial number 39448 because it is 39,448 days after January 1, 1900.
- This function will return the depreciation until the last period of the life of the assets or until the cumulated value of depreciation is greater than the cost of the assets minus the salvage value.
- The life of the asset is calculated by $(1 / \text{rate})$. The depreciation coefficient depends on the life of the asset.
- If the life of the asset is between 3 and 4 years, the coefficient is 1.5.
- If the life of the asset is between 5 and 6 years then the coefficient is 2.
- If the life of the asset is greater than 6 years then the coefficient is 2.5.
- The depreciation rate will grow to 50 percent then the coefficient is 2.5.
- If the life of assets is between 0 (zero) and 1, 1 and 2, 2 and 3, or 4 and 5, the #N/A! error value is returned.

See its example in next lecture

AMORLINC

Returns the depreciation for each accounting period. If an asset is purchased in the middle of the accounting period, the prorated depreciation is taken into account.

Syntax

AMORLINC(cost, date_purchased, first_period, salvage, period, rate, basis)
Cost: cost of the asset

Date_purchased: date of the purchase of the asset.

First_period: date of the end of the first period

Salvage: salvage value at the end of the life of the asset.

Period: period.

Rate: rate of depreciation.

Basis: year basis to be used.

Basis

Date system

0 or omitted 360 days (NASD method)

1 Actual

3 365 days in a year

4 360 days in a year (European method)

AMORLINC-EXAMPLE

	Data	Description
A2	2400	Cost
A3	8/19/2008	Date purchased
A4	12/31/2008	End of the first period
A5	300	Salvage value
A6	1	Period
A7	15%	Depreciation rate
A8	1	Actual basis (see above)

Formula
=ACCRINT(A2,A3,A4,A5,A7,A8)

Result Description
First period interest (30%)

CUMIPMT
Returns the cumulative interest paid between two periods
For description see lecture 8

CUMPRINC
Returns the cumulative principal paid on a loan between two periods
For description see lecture 8

DB
Returns the depreciation of an asset for a specified period using the fixed-declining balance method.

Syntax

DB(cost,salvage,life,period,month)

Cost is the initial cost of the asset.

Salvage is the value at the end of the depreciation (sometimes called the salvage value of the asset).

Life is the number of periods over which the asset is being depreciated (sometimes called the useful life of the asset).

Period is the period for which you want to calculate the depreciation. Period must use the same units as life.

Month is the number of months in the first year. If month is omitted, it is assumed to be 12.

Remarks

- The fixed-declining balance method computes depreciation at a fixed rate. DB uses the following formulas to calculate depreciation for a period:
(cost - total depreciation from prior periods) * rate
where:
 $rate = 1 - ((salvage / cost) ^ (1 / life))$, rounded to three decimal places
- Depreciation for the first and last periods is a special case. For the first period, DB uses this formula:
 $cost * rate * month / 12$
- For the last period, DB uses this formula:
 $((cost - total depreciation from prior periods) * rate * (12 - month)) / 12$

DD

Returns the depreciation of an asset for a specified period using the double-declining balance method or some other method you specify

Syntax

DD(cost,salvage,life,period,factor)

Cost is the initial cost of the asset.

Salvage is the value at the end of the depreciation (sometimes called the salvage value of the asset). This value can be 0.

Life is the number of periods over which the asset is being depreciated (sometimes called the useful life of the asset).

Period is the period for which you want to calculate the depreciation. Period must use the same units as life.

Factor is the rate at which the balance declines. If factor is omitted, it is assumed to be 2 (the double-declining balance method).

Important All five arguments must be positive numbers.

Remarks

- The double-declining balance method computes depreciation at an accelerated rate. Depreciation is highest in the first period and decreases in successive periods. DD uses the following formula to calculate depreciation for a period:
 $Min((cost - total depreciation from prior periods) * (factor / life), (cost - salvage - total depreciation from prior periods))$

- Change factor if you do not want to use the double-declining balance method

MIRR

Returns the modified internal rate of return for a series of periodic cash flows. MIRR considers both the cost of the investment and the interest received on reinvestment of cash.

Syntax

MIRR(values,finance_rate,reinvest_rate)

Values is an array or a reference to cells that contain numbers. These numbers represent a series of payments (negative values) and income (positive values) occurring at regular periods. Values must contain at least one positive value and one negative value to calculate the modified internal rate of return. Otherwise, MIRR returns the #DIV/0! error value.

If an array or reference argument contains text, logical values, or empty cells, those values are ignored; however, cells with the value zero are included.

Finance_rate is the interest rate you pay on the money used in the cash flows.

Reinvest_rate is the interest rate you receive on the cash flows as you reinvest them.

INTERNAL RATE OF RETURN IRR

Returns the internal rate of return for a series of cash flows.

Syntax

IRR(values,guess)

Values is an array or a reference to cells that contain numbers for which you want to calculate the internal rate of return.

- Values must contain at least one positive value and one negative value to calculate the internal rate of return.
- IRR uses the order of values to interpret the order of cash flows. Be sure to enter your payment and income values in the sequence you want.
- If an array or reference argument contains text, logical values, or empty cells, those values are ignored.

Guess is a number that you guess is close to the result of IRR.

- Microsoft Excel uses an iterative technique for calculating IRR. Starting with guess, IRR cycles through the calculation until the result is accurate within 0.00001 percent. If IRR can't find a result that works after 20 tries, the #NUM! error value is returned.
- In most cases you do not need to provide guess for the IRR calculation. If guess is omitted, it is assumed to be 0.1 (10 percent).
- If IRR gives the #NUM! error value, or if the result is not close to what you expected, try again with a different value for guess.

IRR-EXAMPLE

In the slide the Excel worksheet is shown. In cell A97, the investment of 70,000 is entered with minus sign to denote negative cash flow.

In cell A98 to A102, revenue per year (1-4) is entered. In cell A103, only years 1 to 4 were selected from the revenue stream. The IRR is -2% in this case.

In the next formula in cell A105, the entire revenue stream was considered. The IRR improved to 9%.

Next only first 2 years of revenue stream were considered with an initial guess of -10%. The result was -44%.

3M		X ✓ & =SLN(A68:A69,A70)	
	A		B
66	SLN(cost,salvage,life)		
67	Data	Description	
68	30000	Cost	
69	7500	Salvage value	
70	10	Years of useful life	
71	=SLN(A68;A69;A70)		
72	The depreciation allowance for each year (2,250)		
73			

SYD

Returns the sum-of-years' digits depreciation of an asset for a specified period

Syntax

SYD(cost,salvage,life,per)

Cost is the initial cost of the asset.

Salvage is the value at the end of the depreciation (sometimes called the salvage value of the asset).

Life is the number of periods over which the asset is depreciated (sometimes called the useful life of the asset).

Per is the period and must use the same units as life.

Remark

- SYD is calculated as follows:

$$SYD = \frac{(cost - salvage) \cdot (life - per + 1) \cdot 2}{(life)(life + 1)}$$

5YD: cost salvage life per
 3YD= ((Cost-Salvage)/(life-per+1)^2)/(life)/(life+1)
 Data Description
 30000 Initial cost
 7500 Salvage value
 10 Lifespan in years
 =5YD(A80:A81:A82:1)
 Yearly Depreciation allowance for the first year (4,090.91)

VDB

Returns the depreciation of an asset for any period you specify, including partial periods, using the double-declining balance method or some other method you specify. VDB stands for variable declining balance.

Syntax

VDB(cost,salvage,life,start_period,end_period,factor,no_switch)

Cost is the initial cost of the asset.

Salvage is the value at the end of the depreciation (sometimes called the salvage value of the asset).

Life is the number of periods over which the asset is depreciated (sometimes called the useful life of the asset).

Start_period is the starting period for which you want to calculate the depreciation. Start_period must use the same units as life.

End_period is the ending period for which you want to calculate the depreciation. End_period must use the same units as life.

Factor is the rate at which the balance declines. If factor is omitted, it is assumed to be 2 (the double-declining balance method). Change factor if you do not want to use the double-declining balance method. For a description of the double-declining balance method, see DDB.

No_switch is a logical value specifying whether to switch to straight-line depreciation when depreciation is greater than the declining balance calculation.

- If **no_switch** is **TRUE**, Microsoft Excel does not switch to straight-line depreciation even when the depreciation is greater than the declining balance calculation.
- If **no_switch** is **FALSE** or omitted, Excel switches to straight-line depreciation when depreciation is greater than the declining balance calculation.

All arguments except no_switch must be positive numbers.

All arguments except no_switch must be positive numbers.

- VDB: cost salvage life start_period end_period, factor no_switch)
- Data Description
- 2400 Initial cost
- 300 Salvage value
- 10 Lifetime in years
- =VDB(A89;A90;A91*365; A950;1)
- First day's depreciation.
- Excel automatically assumes that factor is 2 (1.32)

IRR
Returns the internal rate of return for a series of cash flows represented by the numbers in values. These cash flows do not have to be even, as they would be for an annuity. However, the cash flows must occur at regular intervals, such as monthly or annually. The internal rate of return is the interest rate received for an investment consisting of payments (negative values) and income (positive values) that occur at regular periods.

Syntax

IRR(values,guess)

Values is an array or a reference to cells that contain numbers for which you want to calculate the internal rate of return.

- **Values** must contain at least one positive value and one negative value to calculate the internal rate of return.
- IRR uses the order of values to interpret the order of cash flows. Be sure to enter your payment and income values in the sequence you want.
- If an array or reference argument contains text, logical values, or empty cells, those values are ignored.

Guess is a number that you guess is close to the result of IRR.

- Microsoft Excel uses an iterative technique for calculating IRR. Starting with guess, IRR cycles through the calculation until the result is accurate within 0.00001 percent. If IRR can't find a result that works after 20 tries, the #NUM! error value is returned.
- In most cases you do not need to provide guess for the IRR calculation. If guess is omitted, it is assumed to be 0.1 (10 percent).
- If IRR gives the #NUM! error value, or if the result is not close to what you expected, try again with a different value for guess.

Remarks

IRR is closely related to NPV, the net present value function. The rate of return calculated by IRR is the interest rate corresponding to a 0 (zero) net present value. The following formula demonstrates how NPV and IRR are related:
NPV(IRR(B1:B6),B1:B6) equals 0.00E+00 (Within the accuracy of the IRR calculation, the value 0.00E+00 is effectively 0 (zero).)

IRR-EXAMPLE

In the slide the Excel worksheet is shown.

- Microsoft Excel stores dates as sequential serial numbers used in calculations. By default, January 1, 1900 is serial number 1. January 1, 2008 is serial number 39448 because it is 39,448 days after January 1, 1900. Microsoft Excel for the Macintosh uses a different date system as its default.
- Numbers in dates are truncated to integers.
- XIRR expects at least one positive cash flow and one negative cash flow; otherwise, XIRR returns the #NUM! error value.
- If any number in dates is not a valid date, XIRR returns the #VALUE! error value.
- If any number in dates precedes the starting date, XIRR returns the #NUM! error value.
- If values and dates contain a different number of values, XIRR returns the #NUM! error value.
- In most cases you do not need to provide guess for the XIRR calculation. If omitted, guess is assumed to be 0.1 (10 percent).
- XIRR is closely related to XNPV, the net present value function. The rate of return calculated by XIRR is the interest rate corresponding to XNPV = 0.
- Excel uses an iterative technique for calculating XIRR. Using a changing rate (starting with guess), XIRR cycles through the calculation until the result is accurate within 0.000001 percent. If XIRR can't find a result that works after 100 tries, the #NUM! error value is returned. The rate is changed until:

where:

$$0 = \sum_{i=1}^n \frac{P_i}{(1 + \text{rate})^{\frac{(d_i - d_1)}{365}}}$$

d_i = the i th, or last, payment date.

d_1 = the 0th payment date.

P_i = the i th, or last, payment.

XIRR EXAMPLE

Here, the investment is in cell A111. The revenue stream is in cells A112 to A115. The dates for each investment or revenue are given in cells B111 to B115. Please note that the dates are in European format year-month-day. On your computer, you may not have this format.

Scientific	01-03-14
Text	01-03-14 5:30
Special	14-3-2001
Custom	Locale (location):
	Swedish

Date formats display date and time serial numbers as date values. Except for times that have an asterisk (*), applied formats do not switch date orders with the operating system.

OK Cancel

In cell A116, the formula =XIRR(A111:A115,B111:B115,0.1), the range A111:A115 is the cost and revenue stream. The range B111:B115 is the stream for dates. The third term 0.1 is the initial guess for XIRR. The answer in fraction or % is given in cell B116(37.34%).

LINEAR EQUATIONS

Linear equations have following applications in Merchandising Mathematics:

- Solve two linear equations with two variables
- Solve problems that require setting up linear equations with two variables
- Perform linear Cost-Volume-Profit and break-even analysis employing:
- The contribution margin approach
- The algebraic approach of solving the cost and revenue functions

SOLVING LINEAR EQUATIONS –

Here is an example of solving simultaneous linear equations.

$$\begin{aligned} 2x - 3y &= -6 \\ x + y &= 2 \end{aligned}$$

Solve for y

$$\begin{aligned} 2x - 3y &= -6 \\ 2x + 2y &= 4 \\ \hline -5y &= -10 \\ y &= 10/5 \\ y &= 2 \end{aligned}$$

Let us look at the same equations again.

$$\begin{aligned} 2x - 3y &= -6 \\ x + y &= 2 \end{aligned}$$

We solved for x

Now let us substitute y by 2

$$\begin{aligned} 2x - 3(2) &= -6 \\ 2x - 6 &= -6 \\ 2x &= 0 \\ x &= 0 \end{aligned}$$

Check your answer

By substituting the values into each of the equations

Equation 1:

$$2x - 3y = -6$$

$$x = 0 \quad y = 2$$

$$\text{LHS} = 2x - 3y = 2(0) - 3(2)$$

$$= -6 = \text{RHS}$$

Equation 2:

$$x + y = 2$$

$$\text{LHS} = x + y = 0 + 2 = 2 = \text{RHS}$$

The right side is equal to left hand side. Hence the answer is correct.

LECTURE 20 PERFORM BREAK-EVEN ANALYSIS EXCEL FUNCTIONS FOR FINANCIAL ANALYSIS

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 18
- MS EXCEL Financial Functions
- Perform Break-Even Analysis

LINEAR EQUATIONS

Zain purchases the same amount of commodity 1 and 2 each week. After price increases from Rs. 1.10 to Rs. 1.15 per item of commodity 1, and from Rs. 0.98 to Rs. 1.14 per item of commodity 2, the weekly bill rose from Rs. 84.40 to Rs. 91.70. How many items of commodity 1 and 2 are purchased each week?

Let x = # of commodity 1

Let y = # of commodity 2

Setting up Linear Equations

Equation 1

$$1.10x + 0.98y = 84.40 \quad (1)$$

Eliminate x in (1) by Dividing both sides by 1.10.

$$(1.10x + 0.98y)/1.10 = 84.40/1.10$$

$$x + 0.8909y = 76.73$$

Equation 2

$$1.15x + 1.14y = 91.7 \quad (2)$$

Eliminate x in (2) by Dividing both sides by 1.15

$$(1.15x + 1.14y)/1.15 = 91.70/1.15$$

$$x + 0.9913y = 79.74$$

Result 1:

$$x + 0.8909y = 76.73 \quad (3)$$

$$x + 0.9913y = 79.74 \quad (4)$$

Next:

Subtract (4) from (3):

Result 2:

$$0.1004y = 3.01$$

$$y = 3.01/0.1004$$

Or

$$y = 29.96 \text{ approximately } 30$$

30 items of commodity 2 are purchased each week.

$$1.10x + 0.98y = 84.40$$

Substitution

Substitute value of y in (1)

Result:

$$1.10x + 0.98(29.96) = 84.40$$

Solve

$$1.10x + 29.36 = 84.40$$

$$1.10x = 84.40 - 29.36$$

$$1.10x = 55.02$$

Result:

$$x = 50.02 \text{ approximately } 50$$

50 items of commodity 1 are purchased each week.

Check your answer
New weekly cost
Commodity 1:

50 × 1.15 = 57.50Rs
 Commodity 2:
 30 × 1.14 = 34.20
 Total cost = 91.70Rs

TERMINOLOGY

There are either Business Costs or Expenses.

Break Even Analysis

Break Even Analysis refers to the calculation to determine how much product a company must sell in order to get break even point (The point at which no profit is made and no losses are incurred) on that product. Break-even analysis provides insight into whether or not revenue from a product or service has the ability to cover the relevant costs of production of that product or service. Managers can use this information in making a wide range of business decisions, including setting prices, preparing competitive bids, and applying for loans.

Cost-Volume-Profit Analysis

Cost-volume-profit (CVP) analysis expands the use of information provided by breakeven analysis. It deals with how profits and costs change, with a change in volume. More specifically, it looks at the effects on profits by changes in such factors as variable costs, fixed costs, selling prices, volume. By studying the relationships of costs, sales, and net income, management is better able to cope with many planning decisions. For example, CVP analysis attempts to answer the following questions: (1) What sales volume is required to break even? (2) What sales volume is necessary in order to earn a desired (target) profit? (3) What profit can be expected on a given sales volume? (4) How would changes in selling price, variable costs, fixed costs, and output affect profits?

Fixed Costs (FC)

Fixed Costs are such costs that do not change if sales increase or decrease e.g. rent, property taxes, some forms of depreciation.

Variable Costs (VC)

Variable costs do change in direct proportion to sales volume e.g. material costs and direct labor costs.

Production Capacity (PC)

It is the number of units a firm can make in a given period.

Break Even Point

Break Even point is a point at which neither a profit nor loss is made. Revenue is exactly equal to costs. Break even point can be expressed as

- ① units
- ② Sales or Rupees (Rs)
- ③ Percent of capacity

BEP in units calculates how many units should be sold to break even. If the product is sold in a quantity greater than this, the firm will make a profit; below this point, a loss.

$$\text{BEP in units} = \frac{\text{Fixed Costs}}{\text{Contribution Margin per unit}}$$

BEP in Rs calculates the revenue that must be obtained to reach break even point.

$$\text{BEP in Rs} = \frac{\text{Fixed Costs}}{\text{Contribution Margin}} \times \text{Net Sales}$$

$$\text{BEP in Rs} = \frac{\text{Fixed costs}}{\text{Contribution Margin per unit}} \times \text{Selling Price per unit}$$

BEP as percent of capacity calculates what percent of production capacity will be utilized to produce the number of units required to reach break even point.

$$\text{BEP as \% of capacity} = \frac{\text{BEP in units}}{\text{Production capacity}} \times 100\%$$

Contribution Margin

Contribution Margin is the Rs. amount that is found by deducting Variable Costs from Sales or revenues and "contributes" to meeting Fixed Costs and making a 'Net Profit'. It can be calculated on a total or per unit basis.

$$\text{Contribution Margin} = \text{Net Sales} - \text{Variable Cost} = S - VC$$

$$\text{Contribution margin per unit} = \text{CM} = \text{Sale price per unit} - \text{Variable cost per unit}$$

Contribution Rate (CR)

$$\text{Contribution rate} = \frac{\text{Contribution Margin}}{\text{Net sales}} \times 100\% = \frac{\text{CM}}{S} \times 100\%$$

$$\text{Contribution rate} = \frac{\text{Contribution Margin per unit}}{\text{Sale price per unit}} \times 100\% = \frac{\text{CM}}{S} \times 100\%$$

A CONTRIBUTION MARGIN STATEMENT

	Rs.	%
Net Sales (Price × # Units Sold)	x	100
Less: Variable Costs	x	x
Contribution Margin	x	x
Less: Fixed Costs	x	x
Net Income	x	x

The **net sales** are calculated by multiplying price per unit with number of units sold.

$$\text{Net Sales} = \text{Sale price per unit} \times \text{number of units sold}$$

This figure is treated as 100%

Next, variable costs are specified and deducted from the Net sales to obtain the Contribution Margin.

Next, Fixed costs are deducted from the contribution Margin. The result is **Net Income**.

$$\text{Net Income} = \text{Contribution Margin} - \text{Fixed Costs}$$

Under the % column, percentage of each item is calculated with respect to the Net Sales.

Here's an example.

Net Sales	462,452 Rs	100%
Less: Variable Costs		

Cost of goods sold	230,934 Rs	50%
Sales Commissions	58,852 Rs	12.7%
Delivery Charges	13,984 Rs	3%
Total Variable Costs	(230934 + 58852 + 13984)	65.7%
	303,770 Rs	
Contribution Margin	(462,452 - 303770)	34.3%
	158,682 Rs	
Less: Fixed Costs		
Advertising	1,850 Rs	0.4%
Depreciation	13,250 Rs	2.8%
Insurance	5,400 Rs	1.2%
Payroll Taxes	8,200 Rs	1.8%
Rent	9,600 Rs	2.1%
Utilities	17,801 Rs	3.8%
Wages	40,000 Rs	8.6%
Total Fixed Costs	(1850+13250+5400+8200 +9600+17801+40000)	
	96,101 Rs	20.8%
Net Income	(158682 - 96101)	
	62,581 Rs	13.5%

SCENARIO 1

A firm is planning to add a new item in its product line. Market research indicates that the new product can be sold at Rs. 50 per unit. Cost analysis provides the following information:

Fixed Costs (FC) per period = Rs. 8640

Variable Costs (VC) = Rs. 30 per unit.

Production Capacity per period = 900 units

How much does the sale of an additional unit of a firm's new product contribute towards increasing its net income?

Formula

Contribution Margin per unit = $CM = S - VC$

Contribution Rate = $CR = CM/S \times 100\%$

***Break Even Point (BEP):**

...In Units (x): $Rs. x = (FC / CM)$

...In Sales Rs.: $Rs. x = (FC / CM) \times S$

...In % of Capacity: $BEP \text{ in Units}/PC \times 100\%$

* At Break Even, Net Profit or Loss = 0

Scenario 1 Summary

Selling price per unit = $S = 50$ Rs

Fixed Costs per period = $FC = Rs. 8640$

Variable Costs = $VC = Rs. 30$ per unit

Production Capacity per period = $PC = 900$ units

Solution of this problem is in the next lecture

LECTURE 21 PERFORM LINEAR COST-VOLUME PROFIT AND BREAK-EVEN ANALYSIS USING THE CONTRIBUTION MARGIN APPROACH

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 18
- Perform Break-Even Analysis
- MS EXCEL Financial Functions

OBJECTIVES

The objectives of the lecture are to learn about:

- Review Lecture 20
- Perform linear cost-volume profit and break-even analysis.
- Using the contribution margin approach

SCENARIO 1

Contribution Margin per unit = $CM = S - VC = 50 - 30 = 20$ Rs.
Contribution rate = $CR = CM/S \times 100\% = 20/50 \times 100\% = 40\%$

Break Even Point:

In Units: $x = FC / CM = 8640/20 = 432$ Units

In Rs. $x = (FC / CM) \times S =$

$(Rs. 8640/Rs. 20) \times Rs. 50 = Rs. 21,600$

% of Capacity = $BEP \text{ in units} / PC \times 100\%$

$= 432 / 900 \times 100\%$

$= 48\%$

Thus by selling more than 432 units of its new product, a firm can make profit.

SCENARIO 2

The Lighting Division of A Lighting Fitting Manufacturer plans to introduce a new street light based on the following accounting information:

FC per period = Rs. 3136

VC = Rs.157 per unit

S= Rs.185 per unit

Production Capacity per period = 320 units

Calculate the break even point (BEP)

— in units

— in rupees

— as a percent of capacity

Break Even Point

In units = FC / CM

$CM = S - VC$

$= Rs. 185 - 157 = Rs. 28$

$$\text{BEP in units} = 3136/28 = 112 \text{ Units}$$

Break Even Point

$$\text{In Rupees} = (\text{FC} / \text{CM}) \times S \\ = (3136/28) \times 185 = 20720 \text{ Rs}$$

Break Even Point

$$\text{as a percent of capacity} = \text{BEP in units} / \text{PC} \times 100\%$$

$$= 112/320 \times 100\% \\ = 35\% \text{ Capacity}$$

SCENARIO 2-1

$$\text{FC} = \text{Rs. } 3136$$

$$\text{VC} = \text{Rs. } 157 \text{ per unit}$$

$$S = \text{Rs. } 185 \text{ per unit}$$

$$\text{Production Capacity} = 320 \text{ units}$$

Determine the BEP as a % of capacity if FC are reduced to Rs. 2688

Formula

$$\text{BEP as a \% of capacity} = \text{BEP in units} / \text{PC} \times 100\%$$

Step 1... Find CM

Step 2... Find BEP in units

Step 3... Find \% of Capacity

Step 1... Find CM per unit

$$S = 185 \text{ Rs per unit}$$

$$\text{VC} = 157 \text{ Rs per unit}$$

$$\text{CM} = S - \text{VC} = 185 - 157 = \text{Rs. } 28$$

Step 2... Find BEP in units

$$\text{BEP in units} = \text{FC} / \text{CM}$$

$$= \text{Rs. } 2688 / \text{Rs. } 28$$

$$= 96 \text{ Units}$$

Step 3... Find \% of Capacity

$$\text{BEP as a \% of capacity} = \text{BEP in units} / \text{PC} \times 100\%$$

$$= 96/320 \times 100\%$$

$$= 30\% \text{ of Capacity}$$

SCENARIO 2-2

$$\text{FC} = \text{Rs. } 3136$$

$$\text{VC} = \text{Rs. } 157 \text{ per unit}$$

$$S = \text{Rs. } 185 \text{ per unit}$$

$$\text{Production Capacity} = 320 \text{ units per period}$$

Determine the BEP as a % of capacity if FC are increased to Rs. 4588, and VC reduced to 80% of S.

$$\text{BEP as a \% of capacity} = \text{BEP in units} / \text{PC} \times 100\%$$

$$\text{New VC} = S \times 80\% = 185 \times 0.8 = \text{Rs. } 148$$

$$\text{New FC} = 4588 \text{ Rs}$$

Step 1... Find CM per unit

$$S = 185 \text{ Rs per unit}$$

$$VC = 148 \text{ Rs per unit}$$

$$CM = S - VC = \text{Rs. } 37$$

Step 2... Find BEP in units

$$\text{BEP in units} = FC/CM$$

$$= \text{Rs. } 4588 / \text{Rs. } 37$$

$$= 124 \text{ Units}$$

Step 3... Find % of Capacity

$$\text{BEP as a \% of capacity} = \text{BEP in units} / PC \times 100\%$$

$$= 124 / 320 \times 100\%$$

$$= 39\% \text{ of Capacity}$$

SCENARIO 2-3

$$FC = \text{Rs. } 3136$$

$$VC = \text{Rs. } 157 \text{ per unit}$$

$$S = \text{Rs. } 185 \text{ per unit}$$

$$\text{Production Capacity} = 320 \text{ units per period}$$

Determine the BEP as a % of capacity if S is reduced to Rs. 171.

$$\text{BEP as a \% of capacity} = \text{BEP in units} / PC \times 100\%$$

Step 1... Find CM per unit

$$S = 171 \text{ Rs per unit}$$

$$VC = 157 \text{ Rs per unit}$$

$$CM = S - VC = \text{Rs. } 14$$

Step 2... Find BEP in units

$$\text{BEP in units} = FC/CM$$

$$= \text{Rs. } 3136 / \text{Rs. } 14$$

$$= 224 \text{ Units}$$

Step 3... Find BEP as a % of Capacity

$$\text{BEP as a \% of capacity} = \text{BEP in units} / PC \times 100\%$$

$$= 224 / 320 \times 100\%$$

$$= 70\% \text{ of Capacity}$$

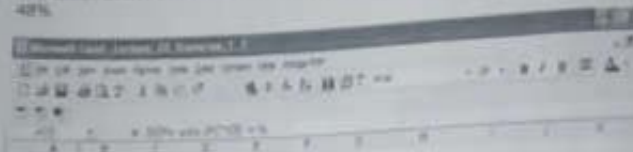
LECTURE 22 PERFORM LINEAR COST-VOLUME PROFIT AND BREAK-EVEN ANALYSIS

OBJECTIVES

- The objectives of the lecture are to learn about:
 - Review Lecture 21
 - Perform Linear Cost-Volume Profit and Break-Even analysis
 - Using Microsoft Excel

SCENARIO 1

Let us look at different scenarios for calculation of contribution margin and net profit. The explanations are given in the slides. The Break-Even Point in Rs. is 21,600. The break-Even point as a % of capacity is 48%.



Scenario 1

Production Capacity	900
Sale = S =	50
Variable Costs = VC =	30
Contribution Margin = CM = S-VC	20 = H4-H5
Fixed Cost = FC =	8640
BEP in Units $x = FC / CM =$	432 = H7/H6
BEP in Rs. $= (FC / CM) * S$	21600 = H8*H4
BEP in units / PC * 100 =	48 = H8/H3*100

SCENARIO 2

The Break-Even Point in Rs. is 20,720. The break-Even Point as a % of capacity is 35%.

$$BEP \% capacity = \frac{BEP \text{ in unit}}{PC} \times 100$$

$$\frac{FC}{CM} / PC$$

Scenario 2

Production Capacity	320
Sale = S =	185
Variable Costs = VC =	157
Contribution Margin = CM = S-VC	28 =H15-H16
Fixed Cost = FC =	3136
BEP in Units $x = FC / CM =$	112 =H18/H17
BEP in Rs. = $(FC / CM) \times S$	20720 =H19*H15
BEP in units /PC*100 %	35 =H19/H14*100

SCENARIO 2-1

The Break Even Point in Rs. is 17,760. The break Even Point as a % of capacity is 30%.

Scenario 2-1

Production Capacity	320
Sale = S =	185
Variable Costs = VC =	157
Contribution Margin = CM = S-VC	28 =H26-H27
Fixed Cost = FC =	2688
BEP in Units $x = FC / CM =$	96 =H29/H28
BEP in Rs. = $(FC / CM) \times S$	17760 =H30*H26
BEP in units /PC*100 %	30 =H30/H25*100

SCENARIO 2-2

The Break Even Point in Rs. is 22,940. The break Even Point as % of capacity is 30%.

Scenario 2-2		
Production Capacity		320
Sale = S =		185
Variable Costs = VC = 0.8 * S =		148
Contribution Margin = CM = S - VC		37 = H41 - H42
Fixed Cost = FC =		4588
BEP in Units $x = FC / CM =$		124 = H44 / H43
BEP in Rs. = $(FC / CM) * S$		22940 = H45 * H41
BEP in units / PC * 100 = %		39 = H45 / H40 * 100

SCENARIO 2-3

The Break Even Point in Rs is 38,304. The break Even Point as % of capacity is 70%.

Scenario 2-3		
Production Capacity		320
Sale = S =		171
Variable Costs = VC =		157
Contribution Margin = CM = S - VC		14 = H53 - H54
Fixed Cost = FC =		3136
BEP in Units $x = FC / CM =$		224 = H56 / H55
BEP in Rs. = $(FC / CM) * S$		38304 = H57 * H53
BEP in units / PC * 100 = %		70 = H57 / H52 * 100

174 Units
Step 2. Find units over BEP
 Number of units above BEP = NI / CM
 $= Rs. 2000 / Rs. 28$
 $= 71 \text{ Units}$
 Total units sold = 71 Units above BEP + 112 BEP Units = 183 units

Scenario 2-6: Problem 2-6	
Q1	Q2
Q3	Q4
Q5	Q6
Q7	Q8
Q9	Q10
Q11	Q12
Q13	Q14
Q15	Q16
Q17	Q18
Q19	Q20
Q21	Q22
Q23	Q24
Q25	Q26
Q27	Q28
Q29	Q30
Q31	Q32
Q33	Q34
Q35	Q36
Q37	Q38
Q39	Q40
Q41	Q42
Q43	Q44
Q45	Q46
Q47	Q48
Q49	Q50
Q51	Q52
Q53	Q54
Q55	Q56
Q57	Q58
Q59	Q60
Q61	Q62
Q63	Q64
Q65	Q66
Q67	Q68
Q69	Q70
Q71	Q72
Q73	Q74
Q75	Q76
Q77	Q78
Q79	Q80
Q81	Q82
Q83	Q84
Q85	Q86
Q87	Q88
Q89	Q90
Q91	Q92
Q93	Q94
Q95	Q96
Q97	Q98
Q99	Q100

Net loss
 Net loss = $NL = \text{Number of units sold below BEP in units} \times \text{Contribution Margin per unit}$

Alternatively,
 Negative sign with Net Income means Net loss. That is
 Net loss = - Net Income
 Number of units sold below BEP in units = - Number of units sold above BEP in units
 Thus
 Net loss = - Net Income = - Number of units sold above BEP in units $\times$ Contribution Margin per unit

SCENARIO 2-6
 FC = Rs. 3136
 VC = Rs. 157
 S = Rs. 185
 Capacity = 320 units
 Find the number of units sold if there is a Net Loss (NL) of Rs. 336?

Formula
 Net loss = Number of units sold below BEP in units $\times$ Contribution Margin per unit
 Number of Units below BEP in units = $(NL) / CM$

Step 2... Find BEP in units

$$\text{BEP in units} = \text{FC} / \text{CM}$$

$$= \text{Rs. } 3136 / \text{Rs. } 28$$

$$= 112 \text{ Units}$$

Step 3... Find units over BEP

$$\text{Number of units above BEP} = \text{NI} / \text{CM}$$

$$= \text{Rs. } 2000 / \text{Rs. } 28$$

$$= 71 \text{ Units}$$

$$\text{Total units sold} = 71 \text{ Units above BEP} + 112 \text{ BEP Units} = 183 \text{ units}$$

Microsoft Excel - Lecture 22 Scenario 3.7										
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Net loss

$$\text{Net loss} = \text{NL} = \text{Number of units sale below BEP in units} \times \text{Contribution Margin per unit}$$

Alternatively,

Negative sign with Net Income means Net loss. That is

$$\text{Net loss} = - \text{Net Income}$$

22-Perform Linear Cost Volume Profit And Break Even Analysis

Step 1... Find CM per unit

S = 185 Rs per unit
 VC = 157 Rs per unit
 CM = 28 Rs
 (CM of Rs.28 per unit)

Step 2... Find BEP in units

BEP in units = FC/CM
 = Rs. 3136/ Rs. 28
 = 112 Units

Step 3... Find units below BEP=NL/CM

Number of unit sales below BEP = Rs. 336/Rs. 28 per Unit
 = 12 Units

Hence:

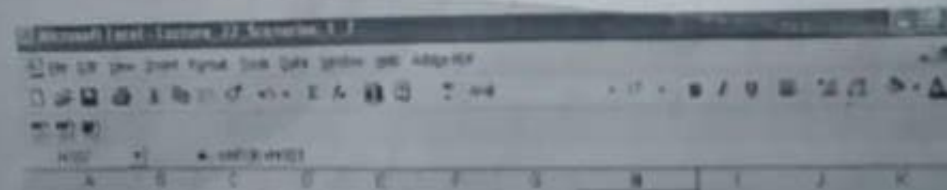
Total Sold Units = BEP in units - Number of units sale below BEP
 = 112 - 12 = 100

Alternate Method

Net loss = - Net income = - Number of units sale above BEP in units $\times$ Contribution Margin per unit

Number of units sale above BEP in units = $\frac{- \text{Net loss}}{\text{Contribution Margin per unit}}$
 = $-336 / 28 = -12$

Total sold units = BEP in units + Number of units sale above BEP in units
 = 112 + (-12) = 100



Scenario 2-6

Production Capacity	320
Sale = S =	185
Variable Costs = VC =	157
Contribution Margin = CM = S-VC	28 =H97-H98
Fixed Cost = FC =	3136
BEP in Units x = FC / CM =	112 =H100/H99
BEP in Rs. = (FC / CM) * S	20720 =H101*H97
BEP in units (PC*100) = %	35 =H101/H96*100

$$\text{Selling Price } S = 1,500 + 0.6S$$

$$S - 0.6S = 1,500 \text{ Rs.}$$

$$\text{Or}$$

$$0.4S = 1,500 = 3,750 \text{ Rs.}$$

Rs. Markup

$$\text{Rs. Markup} = 3,750 \times 0.6$$

$$= 2,250 \text{ Rs.}$$

EXCEL Calculation

Here 1500 is the Purchase price in cell E39.
 % Markup on Sale Price is entered as 60 in cell E40.
 Sale Price was calculated by using the formula $=E39/(1-E40/100)$. The result 3750 is shown in cell D41.
 * EXCEL formula in cell E42 for Rs. Markup was $=E41-E39$. The result 2250 is shown in cell E42.
 Basic formula $S=C+0.6S$ is shown in cell A44. In cell A45 it was simplified to $0.4=C$. In cell A46, it is rewritten as $S=C/0.4S=C/(1-\text{muc})=C/0.4$. Here muc is the %Markup on cost.

Worksheet: Excel - Lecture 14 - Markup, Discount, Markdown					
	A	B	C	D	E
38	SELLING PRICE AGAIN				
39	Purchase price				
40	% Markup on sale price?				
41	Sale price				
42	Rs. Markup?				
43					
44	$S=C+0.6S$				
45	$0.4S=C$				
46	$S=C/0.4S=C/(1-\text{mu})=C/0.4$				

RS. MARKUP AND PERCENT MARKUP ON COST

Tarveer's flower business sells floral arrangements for Rs. 35.
 To make his desired profit, Tarveer needs a 40% Markup on Selling Price.
 What do the flower arrangements Cost Tarveer?
 What is the Rs. Markup?

Selling Price

$$\text{Selling Price} = \text{Cost price} + (\text{Selling price} \times \% \text{Markup on sale})$$

$$* \text{ Net cost price} = \text{List price} - \text{discount price}$$

$$* I = \frac{PRT}{100}$$

* Compound amount

$$S = P \left(1 + \frac{r}{100}\right)^n$$

$$* \text{ Compound interest} = S - P$$

$$* FV = C * \left[\frac{(1+i)^n - 1}{i} \right]$$

$$* PV = C * \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

* Accumulate value

$$S = r \left((1+i)^n - 1 \right) / i$$

OR

= payment for period $\times$ accumulation factor

price

→ formula of discount factor

$$= (1 - 1 / (1 + i)^n) / i$$

→ CUMIPMT

rate nper pv start end type

* CUMPRINC

rate nper pv start end type

→ Effect

Nominal rate, Nper, y.

→ FV Returns Future value of
an investment

5 inputs.

(rate, nper, pmt, pv, type)

* PV

Returns present value of investment

PV (rate, nper, pmt, fv, type)

$$* \text{ Markup cost} = \text{Sale price} - \text{Cost price}$$

$$* \% \text{ markdown} = \frac{\text{mark down}}{\text{cost price}} \times 100$$

$$* \text{ List price} = N = \frac{1}{1 - d\%}$$

$$* \% \text{ profit} = \frac{\text{Profit}}{\text{cost price}} \times 100$$

$$* \text{ Database} = (\text{Salvage cost}, \text{Life period, rents})$$

$$* \text{ Saving} = \text{Income} - \text{cost of expense}$$

$$* \text{ contribution margin} = S - VC \text{ or } \\ = \text{Net sale} - \text{variable cost}$$

$$* \text{ BEP unit} = \frac{\text{Fix cost}}{\text{Net sale} - \text{variable cost}}$$

$$* \text{ Series trade discount} = \frac{\text{Net price}}{\text{List price} (1 - d\%)}$$

$$* \text{ Compound interest} = \text{Sale} - \text{percentage}$$

$$S = P (1 + R\%)^n$$

$$\% = B \times R$$

* Amount of discount =
% discount $\times$ List price

*

REMEMBER IN PRAYERS
KEEP SMILING
SAY NO TO CURRUPTION
HELP TO OTHERS

Regards,
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